

Numerical assessment of plasticity development and energy expenditure of ant-like microtunnelling

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ABSTRACT

This study aims to investigate the potential mechanical benefits of the excavation technique adopted by Harvester ants, that dig tunnels in multiple passes, progressively augmenting the diameter of their shafts along the entire tunnel length during each pass. Bio-inspired abstraction is employed to extract the governing principles of ant tunnelling, characterized by progressive tunnel diameter augmentation, and a proof-of-concept study is conducted to establish a foundation for the development of deployable engineering designs. The analysis is based on circular unsupported tunnels of 10 cm diameter excavated in moist sand deposits. The results can be directly applicable to small exploration robots and microtunnelling devices while also providing insight for larger scale unsupported tunnels in other soil types. The geometry of the excavation-induced plastic zone and the required excavation force, torque, and energy are analysed for a single-pass and a five-stage excavation process. Both excavation techniques are simulated in 2D with the Finite Element Method (FEM). Stress magnitudes calculated with the FEM are introduced in analytical expressions to evaluate the total jacking force, torque, and energy required to excavate an elementary length of tunnel. Over 640 combinations of soil elasto-plastic parameters and cavity depth-to-diameter ratios are analysed. The geometry of the plastic zone that develops over successive excavation passes exhibits a higher degree of complexity than during a single-pass excavation, particularly at greater depth — as reflected by fractal dimensions and Minkowski functionals. Staged excavations lead to more localized plastic deformation zones for soils with low apparent cohesion, high depth-to-diameter ratio, high internal friction angle, and low dilation angle. Emulating the staged excavation process of Harvester ants can reduce the force, torque, energy and power that are needed for microtunnelling compared to more traditional single-pass excavation processes, particularly for low apparent cohesion soils and deep cavities. The key to power and energy savings lies in the incremental formation of localized plastic zones around the tunnel face, which is accompanied by stress redistribution and arching. This lowers the lateral earth pressure that must be supported at the excavation front, the normal stress that controls the shear strength against the applied torque, and the radial stress normal to the sides of the excavating device which controls the frictional force that must be overcome to advance the tunnelling device.

1. Introduction

Ants are remarkably efficient excavators whose energy consumption per unit of volume excavated is 2–4 orders of magnitude less than mechanized tunnelling systems such as Tunnel Boring Machines (TBM) (Soga, 2011). Florida harvester ants typically construct their nests in loose unsaturated fine sands, where capillarity provides additional stability (Espinoza and Santamarina, 2010). Their nests feature helicoidal shafts and ellipsoidal chambers (Tschinkel, 2004) – geometries that have been shown to induce arching effects and promote mechanical stability (Frost et al., 2017; Belachew et al., 2024). Toffin

et al. (2009) suggested that ants burrow their tunnels by progressively augmenting the diameter of their shafts through multiple passes to avoid congestion. Time-lapse photography from 2D laboratory experiments by Yamamoto et al. (2019) further revealed that Harvester ants excavate by first burrowing narrow tunnels and subsequently enlarging their cross-sections in a sequential manner. The schematic presented in Fig. 1 illustrates how ants initially excavate a small-diameter tunnel along a shaft segment that is long relative to its cross-sectional dimensions, and subsequently widen the tunnel through repeated back-and-forth passes. These observations raise a fundamental question:

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Nomenclature			
D_{50}	Median particle size	A_{face}	Tunnel face area
d	Tunnel diameter	A_{side}	Area at tunnelling machine side and soil interface
r	Tunnel radius	OR	Opening ratio
λ	Relaxation factor	μ	Interface friction coefficient
σ_R	Radial stress	CAR	Cutting area ratio
σ_0	In-situ radial stress	F_1	Force to support lateral earth pressure at face
σ_v	Vertical stress	F_2	Force needed to overcome side friction
σ_h	Horizontal stress	F_T	Total thrust force
K_0	Coefficient of lateral earth pressure at rest	T	Excavation torque
σ_θ	Ortho-radial stress	W_T	Total excavation work
u_R	Radial displacement	δ	Advance per revolution
G	Shear modulus	r	Pearson correlation
D	Tunnel depth	x_i, y_i	Paired observations for variables X, Y
γ	Unit weight	$\bar{x}, \bar{y}$	Sample means of observations in variables X, Y
E	Young's modulus	r_s	Spearman correlation
ν	Poisson's ratio	$R_{X,i}, R_{Y,i}$	Paired rank of observations within variables X, Y
c'	Apparent cohesion	$\bar{R}_{X,i}, \bar{R}_{Y,i}$	Sample means of observation ranks within variables X, Y
ϕ	Friction angle	$f_h(x)$	Kernel-density estimation(KDE)
ψ	Dilation angle	$K(u)$	Kernel function
$\epsilon_{s,p}$	Plastic shear strain	h	Bandwidth of KDE or smoothing parameter
$(\epsilon_{s,p})_{\text{max}}$	Max plastic shear strain	n	Sample size
$(\epsilon_{s,p})_{\text{mean}}$	Mean plastic shear strain	d	Number of dimensions
$CV_{\epsilon_{s,p}}$	Coefficient of variation	σ	Sample standard deviation
D_b	Box-counting fractal dimension	y_i	Prediction for the i th sample from additive decomposition
D_{DBC}	Differential box-counting (DBC) fractal dimension	y_{base}	SHAP baseline output
ϵ	Box size (width and length)	$f(x_i, j)$	SHAP value
$N(\epsilon)$	Number of boxes at box size ϵ	$\hat{y}_i$	Prediction for the i th sample from additive ensemble
M	Size of cropped square image	$\mathbf{x}_i$	M-dimensional of feature vector
$n_e(i, j)$	Number of boxes at the (i, j) th cell in DBC	K	Number of trees in the ensemble
$g_{\text{max}}, g_{\text{min}}$	Maximum and minimum grey levels at the (i, j) th cell in DBC	$\mathcal{F}$	Function space of regression trees
G_g	Total number of grey levels in the image	$\tilde{\mathcal{L}}^{(t)}$	XGBoost objective at iteration t
h	Box height	γ_t	Penalty coefficient
N_ϵ	Total DBC boxes	Ω	Tree-complexity regularizer
M_0, M_1, M_2	Minkowski functionals	T_t	Number of leaves in tree t
X	Plastic zone contour	λ_r	L_2 regularization coefficient
R	Local radius of curvature	f_t	Regression tree added at iteration t
S	Surface area of plastic zone	w	Leaf weights
C	Boundary length of plastic zone		
χ	Euler characteristic		
$\beta_0, \beta_1, \beta_2$	Betti numbers		
σ_n	Normal stress at face		

Does sequential enlargement of shallow tunnels in loose unsaturated sand offer mechanical benefits compared to single-pass excavation?

To address this question, this numerical study aims to assess whether or not burrowing a tunnel in sand via successive cavity enlargements yields less plasticity and requires less energy than via a single-pass excavation. The insights from the analysis are meant to be applicable for small diameter unsupported tunnels in loose unsaturated sands or larger scale unsupported tunnels in more cohesive deposits. More than 56% of the world's population lives in urban areas today, and this number is expected to reach 70% by 2050 (Satterthwaite, 2020; World Bank, 2025). The rapid population growth in cities is accompanied by a continuous need to upgrade subsurface infrastructure, particularly utilities. Microtunnelling is often used to install underground pipelines with high precision, minimal surface impact, and in conditions where open excavation is not feasible. Geological variability, particularly at shallow depth, can reduce the safety and efficiency of microtunnelling operations. That is why microtunnel modelling and optimization are active research areas in geotechnical engineering (Moharrami et al., 2022; Leu et al., 2025).

Complex elasto-plastic stress paths involved in sequential excavation methods limit the relevance of analytical solutions to assess the mechanical performance of ant-like burrowing processes. Experimental measurements of stability and energy-related metrics are challenging, and no standardized protocol exists for this type of investigation. Numerical methods provide viable alternatives.

Simulations conducted with the Discrete Element Method (DEM) have examined how soil arching influences the stability of cavities of various sizes and locations in granular media (Frost et al., 2017), showing that ants exploit arching and other soil micro-mechanisms to achieve stable configurations. Coupling level-set DEM with 3D X-ray computed tomography (XRCT) has allowed simulation of various grain-scale processes and visualization of force chains (Buarque de Macedo et al., 2021), which revealed that ants preferentially remove low-stress particles and leverage stress redistribution and relaxation via granular arching. At a larger scale, a DEM model coupled with a model based on the Finite Element Method (FEM) has allowed simulation of sequential excavation under different softening paths and validated new softening laws (Li and Liu, 2024). A DEM model coupled with the

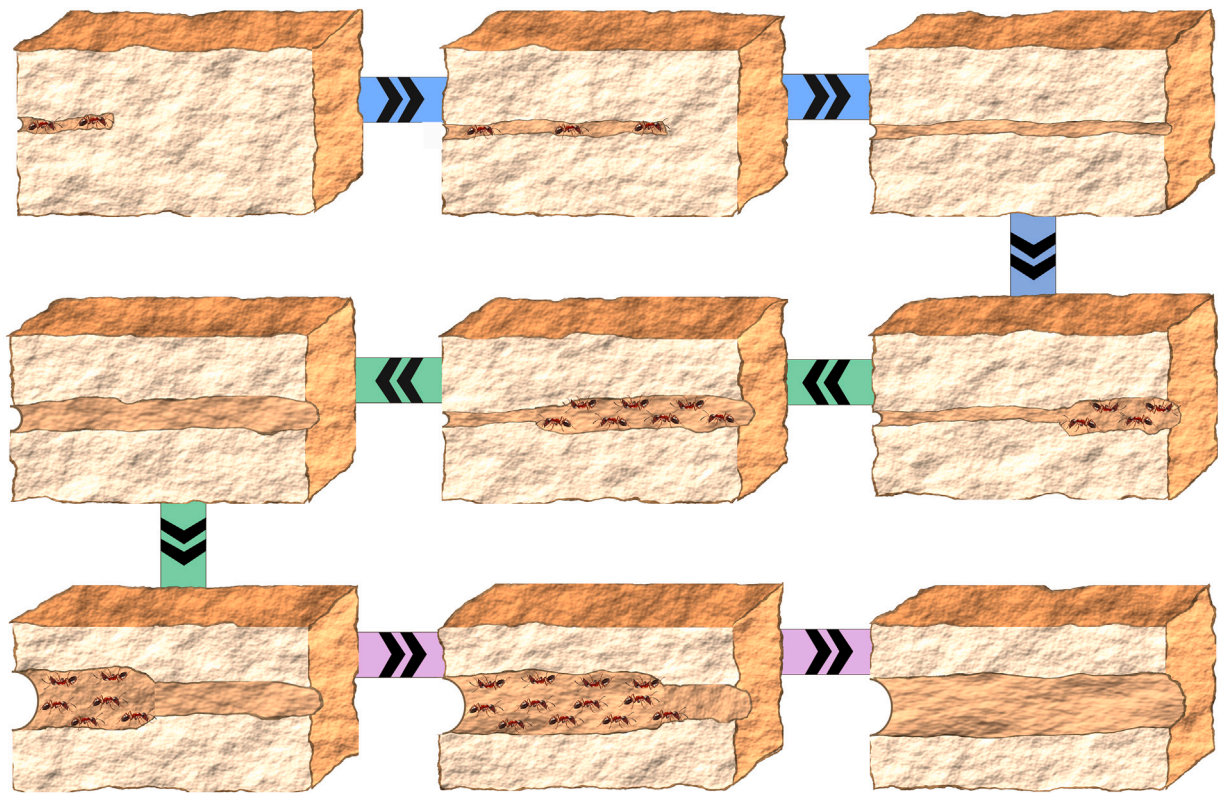


Fig. 1. Schematic representation of sequential excavation by ants.

Finite Difference Method (FDM) has introduced representative-element-volume equivalent parameters, whereby the DEM is used to resolve the excavation-disturbed zone while the surrounding ground in the far field is modelled with the FDM (Zhou et al., 2024). The DEM has also produced stability charts for shallow, small-diameter microtunnels and drill holes in lunar regolith and reduced gravity (Pelech et al., 2022).

The DEM is particularly powerful for capturing particle-scale mechanics but is computationally intensive, making it impractical for large-scale parametric studies. The FEM, in contrast, is computationally more efficient and widely applicable in geomechanics, provided that the scale of the boundary value problem is sufficiently large relative to particle size, thereby justifying a continuum approximation (see Section 3). 3D FEM models of Florida harvester ant nests have shown that the helical structure of the main shafts promotes arching effects, distributes shear stress more evenly and reduces the risk of tensile failure (Belachew et al., 2024). At a larger scale, different excavation sequences have been evaluated with 3D FEM models to identify optimal schemes for tunnelling in urban areas (Sharifzadeh et al., 2013), where the excavation face is often partitioned into temporary drifts to enhance stability and limit surface deformation through the sequential excavation method. A robust 3D FEM model has also been used to study microtunnelling-induced pavement settlement under traffic load (Alzabeebee and Keawsawasvong, 2023). Furthermore, the FEM has been used extensively in the past few years to generate synthetic datasets for training surrogate models. For example, Machine Learning and intelligent optimization algorithms have been used to invert geological parameters and grouting variables, and predict surface settlement without conducting numerous finite element analyses (Liu et al., 2024). A Physics-Informed Neural Network (PINN) framework has also allowed prediction of displacement and stress fields in pressurized cavity expansion problems that involve homogeneous and heterogeneous isotropic linear elastic host media, with arbitrary boundary conditions and constitutive parameters (Liu and Arson, 2025, 2026).

In this study, both the staged and single-pass excavation techniques are simulated in 2D with the FEM for a circular opening of nominal diameter 10 cm. FEM results are analysed statistically to assess the effect of the excavation method on the development of a plastic zone around the tunnel cross-section and to calculate the work input needed to remove soil mass and excavate a finite length of tunnel. Section 2 defines the concept of bio-inspired design and explains how the findings of this study can be used for technology transfer. Section 3 summarizes the methods used to generate data with the FEM and to analyse the results. Section 4 describes the correlations established between the plastic zone and energy cost metrics on the one hand, and the soil constitutive parameters and cavity depth-to-diameter ratio on the other hand. Section 4 also discusses the effects of the excavation method on the distribution of the plastic zone and on energy cost metrics, and the importance of each input parameter for each output metric. Section 5 summarizes the findings and highlights the most relevant implications for engineering practice.

2. Bio-inspired design and technology readiness framework

Biologically inspired design (BID) is an interdisciplinary approach to problems that seeks to transfer principles and solutions that exist in nature into engineering solutions through abstraction, analogy, and functional transfer. This effort is driven by the growing need for environmentally responsible, sustainable, and resilient engineering designs as population growth, urban densification, and climate change increasingly strain existing infrastructure systems. Beyond merely mimicking biological systems, BID emphasizes the identification and study of underlying mechanisms that enable biological systems to achieve robust, efficient, and adaptive performance under complex environmental constraints (Helms et al., 2009; Helms, 2013; Shu et al., 2011). This abstraction-driven perspective is particularly important because biological and engineering contexts differ substantially in scales and mechanisms.

2.1. Levels of abstraction in BID

BID processes are commonly categorized as either problem-driven or solution-driven, both of which consist of iterative abstraction and interpretation across disciplines (Helms et al., 2009; Gallo, 2018). In the problem driven approach, one starts with a specific engineering problem and seeks biological systems with a function that could efficiently solve the engineering problem. In the solution-driven approach, the study of a biological system comes first, and the insights are then applied to engineering problems that could potentially benefit from the function that the biological system performs. Central to both approaches is the extraction of underlying principles from biological systems, which enables analogical transfer beyond the specific biological context in which those principles evolved. This emphasis on abstraction allows biological inspiration to inform engineering design even when direct physical imitation is not feasible. To enable technology transfer across scales and mechanisms from biology to engineering, systematic classification of biological strategies by function is required (Vincent, Julian and Mann, 2002). By framing biological systems in terms of the functions they perform, engineers can identify shared areas across biological and engineered systems. Such functional abstraction is especially relevant for geotechnical engineering problems such as the study presented here, where the governing mechanisms at the engineering scale differs from those under which biological systems evolved.

2.2. Functional decomposition and engineering translation

Functional decomposition provides a structured method for implementing abstraction in bio-inspired design. By decomposing a system into a hierarchy of functions and sub-functions, complex behaviours can be analysed independently from any contextual restriction, and transferred to generalized systems (Sidorenko et al., 2021; Hernandez et al., 2024). Within BID, functional decomposition maps biological systems to engineering objectives by identifying the functions that contribute to the desired engineering-scale design. In some cases, the functions identified from different biological systems are compounded to form a more robust system-level solution. Functional decomposition is particularly effective for linking behaviours observed in biological systems and desired design solutions at the process or component level (Hernandez et al., 2024). In this context, biological systems are not treated as specific templates, but rather as sources of transferable functional strategies. This approach aligns with the present study, which focuses on evaluating generalized excavation-related mechanisms — such as staged material removal, stress redistribution, and energy efficiency — rather than mimicking the exact scale and mechanism of the biological analog (Harvester ants).

2.3. Technology Readiness Level positioning

The Technology Readiness Level (TRL) framework provides a widely adopted means for contextualizing the maturity of emerging technologies, ranging from basic scientific observation (TRL 1) to full commercial deployment (TRL 9) (Kimmel et al., 2020; Olechowski et al., 2015). Early TRLs (TRL 1–3) are characterized by fundamental understanding, conceptual formulation, and proof-of-concept validation, whereas higher TRLs emphasize prototyping, system integration, and operational demonstration.

The present work is intentionally situated within TRL 1–3, where the objective is to establish mechanistic feasibility rather than technological implementation. By employing numerical simulations to assess the mechanical and energetic implications of biologically inspired excavation strategies, this study contributes to the abstraction and proof-of-concept stages of bio-inspired underground engineering. Such early-stage investigation is critical, as premature progression towards higher TRLs without adequate abstraction and validation has been

a recurring challenge in technology development (Olechowski et al., 2015). Therefore, this work represents an early step in the bio-inspired design pipeline, providing the fundamental understanding and validated conceptual basis necessary to inform future experimental and field-scale investigations. Specifically, the function of staged excavation of an unsupported cavity is abstracted from ant tunnel excavation, and its potential benefits in terms of stability and excavation energy expenditures are evaluated for future scale-up and implementation in tunnelling practice.

3. Methods

The methodology in this study involves three steps which are the FEM modelling, post-processing of model outputs and statistical analysis of results. FEM simulations include verification of the selected modelling approach and parametric studies. The second step involves reducing the individual FEM simulation results into scalar metrics that describe the post excavation state and estimated energy expenditure. The third and final step involves assessing the sensitivity of the scalar output metrics and ranking the importance of the input model parameters to the prediction of the most meaningful outputs. The results of the three step process is then used to compare the output metrics of the two excavation methods in light of the important model parameters. A flow chart demonstrating the entire process is shown in Fig. 2.

3.1. Finite element modelling

3.1.1. Numerical modelling approach

This study draws inspiration from the excavation behaviour of Harvester ants, that excavate their nests by progressively augmenting the diameter of their shafts through multiple passes along the entire tunnel length, which contrasts with human construction processes in which mass is typically removed throughout the entire cross section of the cavity being excavated. Some approaches currently used in engineering practice employ staged excavation, including the heading-and-bench method—also referred to as the New Austrian Tunnelling Method (NATM) or Sequential Excavation Method (SEM)—in which the upper portion of the tunnel cross-section (heading) is excavated first, followed by the lower portion (bench) (see Rabcewicz (1964)). This staged advance promotes localized stress redistribution and controlled deformation near the excavation face, thereby enhancing short-term stability in weak or deformable ground. However, in conventional heading-and-bench excavation, stress redistribution is primarily restricted to the immediate vicinity of the excavation front and is rapidly constrained by the installation of ground support. As a result, the surrounding ground does not undergo the broader, unconstrained stress redistribution observed in ant-like excavation, where the tunnel diameter augmentation occurs progressively in multiple passes along the entire length of the tunnel and without structural support. According to field and laboratory observations (Belachew et al., 2024), the median particle diameter (D_{50}) of soils in regions where Harvester ants excavate is in the order of 0.3 mm, while the laboratory observations that inspired this study were done on sand of $D_{50} = 0.1$ mm. The typical diameter (d) of Harvester ant nest shafts is about 10 mm. With a ratio d/D_{50} above 30, a continuum soil model is appropriate to simulate the domain around the ant nest while discrete approaches such as the DEM are better suited for much lower d/D_{50} ratios, as discussed in Davis et al. (1980) and Espinoza and Santamarina (2010). In the following, FEM simulations are conducted to assess the irreversible deformation induced by, and work input required for shaft excavation through two methods: (i) mass removal in one stage, as typically done in engineering practice, and (ii) staged mass removal, as observed in Harvester ant colonies.

Nest excavation is a three-dimensional problem. Excavation of long shafts can usually be modelled in plane strain, unless the advancement front is close to the section considered, in which case, the stress and

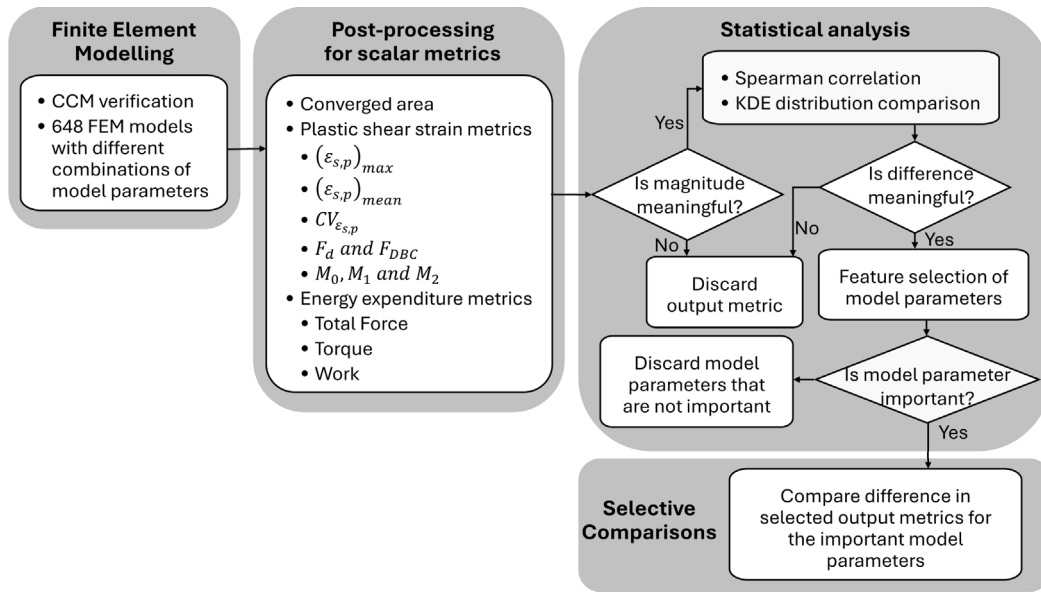


Fig. 2. Process used to analyse the difference between the two excavation schemes. CCM: Convergence-Confinement Method. FEM: Finite Element Method.

displacement fields can only be calculated with a 3D model. The Convergence Confinement Method (CCM) (Panet et al., 2001) was designed to avoid computationally intensive 3D simulations, and is used for tunnel analysis and design. In the CCM, a confinement loss parameter (also called relaxation factor) λ is introduced to simulate the progressive loss of pressure in the rock or soil mass around the tunnel in 2D, as the tunnel face advances through the 2D cross-section of interest (see Fig. 3). The relaxation factor λ is defined as the ratio of the current radial stress at the cavity wall (σ_R) to the in-situ radial stress before excavation (σ_0), often approximated by the overburden pressure. At the onset of excavation, $\lambda = 1$, reflecting the full confinement provided by the geological medium prior to excavation. As excavation advances and the ground begins to deform, the radial stress at the tunnel boundary decreases. For a completely unsupported cavity, the confinement is fully lost, and λ decreases to zero. In supported systems (e.g., shotcrete lining or segmental lining), the relaxation process is halted earlier, resulting in a value $0 < \lambda < 1$ that depends on the stiffness of the support, and on the length of the non-supported section of the tunnel between the face and the cross-section under study. If the formation exhibits a time-dependent behaviour, the time elapsed between excavation and support placement also affects the ultimate value of λ .

As a verification, the progressive excavation of a tunnel of circular cross-section was simulated with a 3D FEM model, and stress and deformation components at the crown and waist points of the cavity wall were compared to those obtained with a 2D FEM model in which the stress at the cavity wall is calculated via the relaxation factor λ . These simulations were conducted using ABAQUS to conveniently extract the time history along the loading paths of the tunnel crown and waist points used for comparison as shown in Fig. 5. In the 3D FEM model, solid mass is removed in sections to simulate the progression of the tunnel excavation front in a soil domain. In the 2D FEM model, the advancement of the excavation front in the direction perpendicular to the plane of analysis is simulated by decreasing the radial stress at the cavity wall from the in-situ stress ($\sigma_R = \lambda \times \sigma_0$, $\lambda = 1$) to a zero radial stress ($\sigma_R = 0$, $\lambda = 0$). To avoid boundary effects, the size of the soil domain is set to $10d$, where d is the tunnel diameter (with $d = 10$ cm). The lateral sides of the domain are subjected to zero normal displacement and zero shear stress, and the three translation degrees of freedom at the bottom face are fixed to zero. The top surface is subjected to a zero shear stress and to a normal compression stress equivalent to an overburden pressure for a tunnel at a depth of

6d. The soil parameters used in these verification simulations are the following: soil density $\gamma = 18$ kN/m³, Young's modulus $E = 40$ MPa, Poisson's ratio $\nu = 0.25$, apparent cohesion $c' = 1.5$ kPa, friction angle $\phi = 28^\circ$ and dilation angle $\psi = 5^\circ$. These soil properties are within the range of values explored in this study (as described in Section 3.1.2) and were thus considered representative for the model verification exercise. Fig. 4 shows the boundary conditions adopted in the 3D and 2D simulations, along with the plastic shear strain fields obtained with the two modelling approaches after full deconfinement. Fig. 5 compares the displacement and stress evolutions at the crown and waist of a circular cavity during the excavation process, simulated in 2D vs. 3D. The ground reaction curve (Fig. 5a), commonly used in CCM analysis (Lee et al., 2021), shows that the results obtained with the 2D and 3D FEM models closely match. At the crown $\lambda = \sigma_R/\sigma_0 \approx \sigma_R/\sigma_v$ starts from 1 and reduces to 0 as the excavation progresses. Similarly at the waist, $\lambda = \sigma_R/\sigma_0 \approx \sigma_R/\sigma_h = \sigma_R/(K_0\sigma_v)$ starts from 1 and reduces to 0 and the radial displacement increases. Fig. 5b shows that the stress path curves are obtained in 2D and 3D are in good agreement. In both of the FEM simulations, the ortho-radial stress at the crown remains almost constant during the excavation, with only a slight decrease towards the end of the excavation, as λ goes to 0. At the waist, the ortho-radial stress increases as the radial stress starts to decrease from its initial value of $0.5\sigma_v$. The ortho-radial stress reaches a peak and drops, which is indicative of the initiation of plastic yield. Fig. 4 shows the equivalent plastic shear strains of the zone around the cavity at the end of the excavation. The accumulation of plastic strain releases the ortho-radial stress, which decreases after the peak value to a value close to that at the crown when the deconfinement is complete.

The slight differences between the 2D and 3D results in the plots of Fig. 5 can be explained by the simplifying assumption of the CCM, which assumes that the tunnel can be modelled in plane strain. This assumption leads to inaccuracies for stress distributions close to the excavation front, i.e., for low deconfinement ratios. Overall, the 2D and 3D methods are judged in sufficient agreement to support a 2D tunnel excavation analysis. Furthermore, tunnel diameter augmentation occurs progressively in multiple passes along the entire length of the tunnel with full stress redistribution occurring at each stage. Therefore, the use CCM to model the stress redistribution of each stage in plane strain is justified for the following analysis.

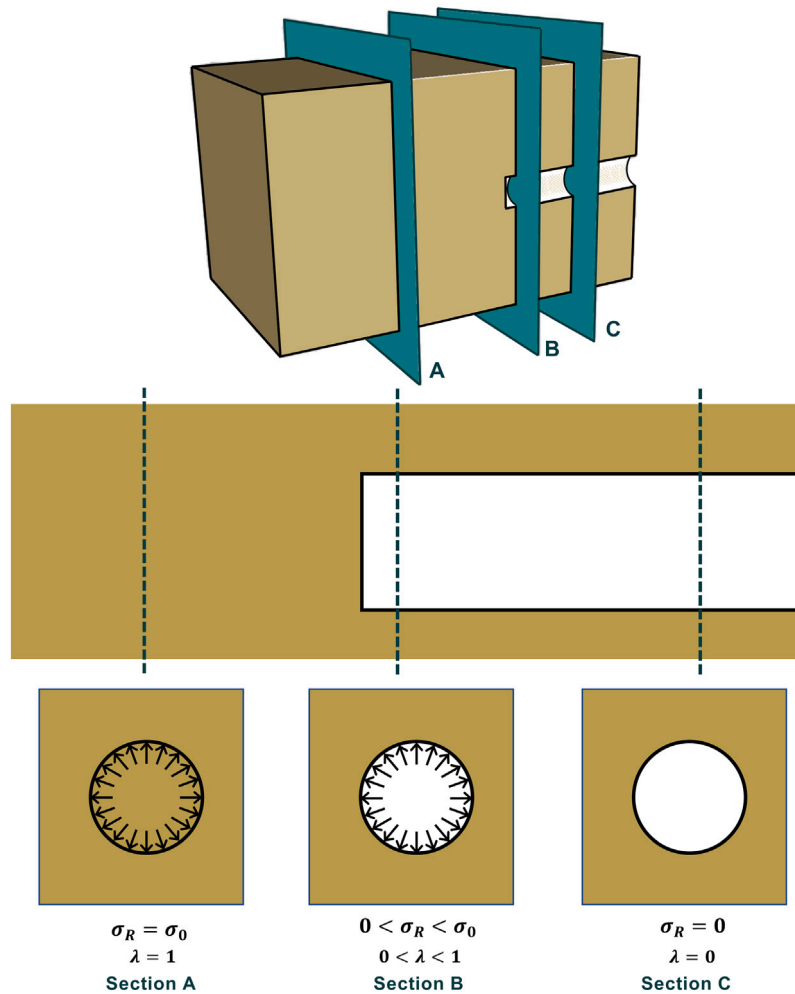


Fig. 3. Principle of confinement loss in the Convergence-Confinement Method. σ_0 and σ_R represent the radial stress at the cavity wall, before excavation and after excavation, respectively.

3.1.2. Specifications of the finite element models

In the proposed FEM model of staged excavation, the ant burrowing technique is mimicked by simulating the excavation of concentric circular cavities of increasing diameter. Fig. 6 depicts the geometry of the Finite Element domain at each simulation step for the single-pass vs. staged excavation processes.

Single-pass and staged mass removal processes are simulated in 2D with the software OptumG2, which is endowed with a built-in application for simulating excavation with the CCM. The displacement and stress fields are extracted at the last step of every excavation stage. Five stages are used for the staged excavation simulations, and only one stage is used for the single-pass excavation cases. Each excavation stage is simulated by decreasing the CCM confinement parameter from 1 (full confinement, no mass removal) to 0 (full deconfinement, excavation completed)

In geotechnical engineering, tunnel diameters can range from 25 mm in mini horizontal directional drilling techniques used for conduit installation (see Trenchless Solutions Ltd. (2025)) to over 20 m in road and rail transportation systems (see California Department of Transportation (2025)). In soft soils such as loose sandy soil, large diameter tunnels need lining support to ensure stability. However, unsupported tunnels of various diameters are possible in cohesive soils and unsaturated sands that have a small amount of apparent cohesion. Ants take advantage of the capillarity in unsaturated sands — which results in apparent cohesion — to keep their tunnels stable without lining support (see Espinoza and Santamarina (2010)). A numerical

parametric study (not presented here) showed that an unsupported tunnel with a diameter of 10 cm could remain stable under the most critical combination of parameters — those yielding the highest stresses and lowest strength — among the cases presented in Table 1. Hence, data for this analysis are generated by varying the tunnel depth-to-diameter ratio over a range informed by documented case histories of microtunnelling projects (see Staheli (2006)), while fixing the tunnel diameter to 10 cm.

As mentioned before, Harvester ants typically build their nests in loose unsaturated sand. An elastic perfectly plastic Mohr–Coulomb model is used to represent the constitutive behaviour of the medium around the cavity, in which the effect of capillarity is accounted for via an apparent cohesion parameter. Data for this analysis are generated by conducting a parametric study in which sand constitutive parameters are varied within ranges of values reported in literature for soils with the aforementioned description (Budhu, 2011; Bolton, 1986; Bowles, 1996; Lambe and Whitman, 1969; Terzaghi et al., 1996). In total, 648 combinations of parameters, shown in Table 1, were studied to understand whether a staged mass removal strategy affects tunnel stability or excavation work input differently than a single-pass mass removal.

3.2. Postprocessing and data generation

The FEM results from OptumG2 are embedded in a complex text-based file that is encoded to allow visualization within the software's

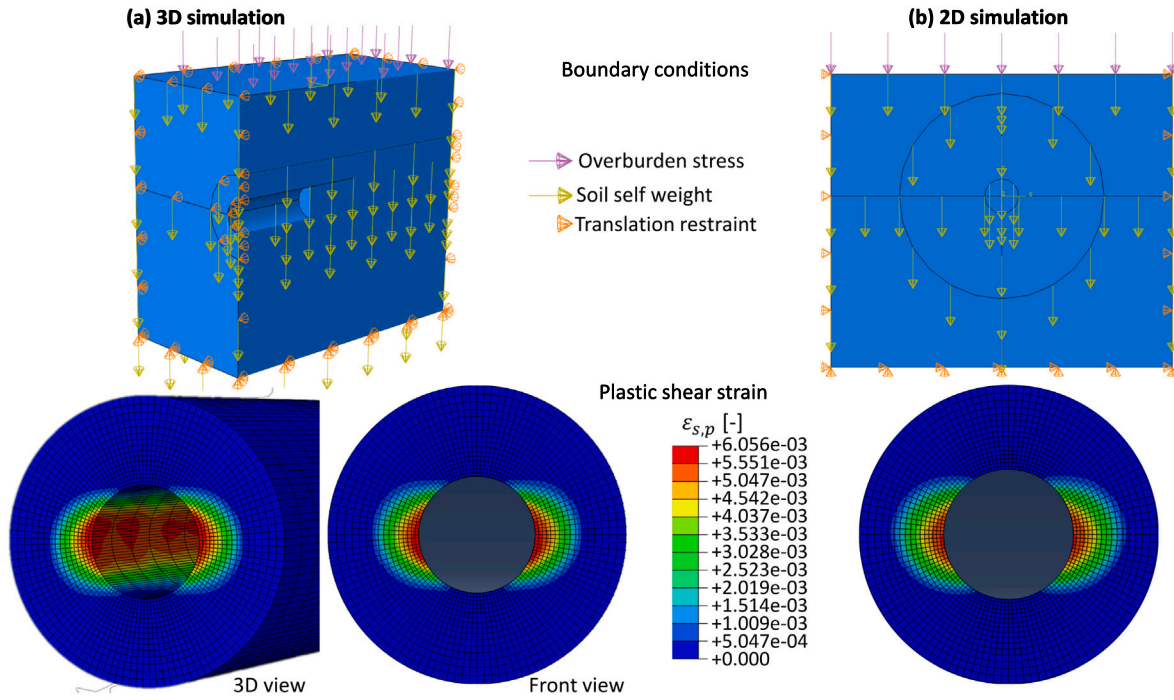


Fig. 4. Results of tunnel excavation simulations in 3D (a) and 2D (b): boundary conditions adopted, and contours of the equivalent plastic shear strains after full deconfinement.

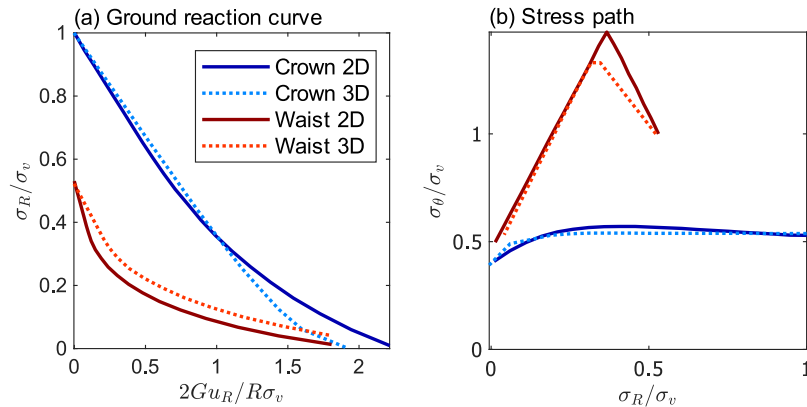


Fig. 5. Comparison of 2D and 3D excavation simulation results at the crown and waist of a circular cavity, where σ_R , σ_θ and σ_v are the radial, ortho-radial and vertical stresses, u_R is the radial displacement (convergence), R is the cavity radius, and G is the shear modulus of the formation. (a) Ground reaction curve (normalized radial stress vs. normalized radial deformation). (b) Stress path (normalized ortho-radial stress vs. normalized radial stress).

Table 1

Physical and constitutive model parameters used in the FEM simulations.

Description	Symbol	Values	Units
Depth-to-diameter ratio	D/d	2, 6, 10	–
Bulk unit weight	γ	16, 20	kN/m ³
Elastic modulus	E	10, 40, 70	MPa
Poisson's ratio	ν	0.25, 0.35	–
Apparent cohesion	c'	1, 5.5, 10	kPa
Soil friction angle	ϕ	28, 33, 38	°
Dilation angle	ψ	5, 25	% of ϕ

interface, but cannot be readily exported in a csv format. Therefore, the text of this file was processed using MATLAB to extract the FEM output field variables at each node of the FEM domain at each step of the analysis. In order to evaluate the difference between the two excavation methods, we used two sets of outputs. The first set is related to the

post-excavation state of the soil domain around the cavity in terms of tunnel convergence and plastic zone characteristics. The second set of metrics aims to estimate the mechanical energy needed to complete the excavation.

3.2.1. Post-excavation state metrics

Convergence. Inward radial displacement at the cavity wall (called convergence) changes the area of the cross-section of the cavity. The change of cross-section area from the excavated cross-section area to the area after convergence, illustrated in Fig. 7, is called “converged area” in the following. To characterize the post-excavation state, we calculated the ratio of the converged area to the excavated area, in percentage. To do this, convergence at the cavity wall was extracted from the displacement field calculated by the FEM.

Plastic strain measures. In the Mohr–Coulomb constitutive model, plasticity is controlled by shear stresses and strains. That is why the

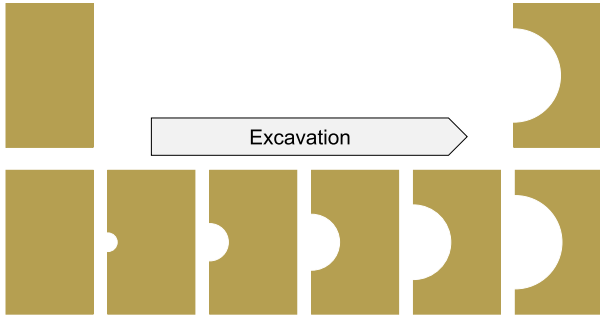


Fig. 6. Schematic representation of the geometry of the Finite Element domain at each simulation step for the single-pass (top) vs. staged (bottom) excavation processes.

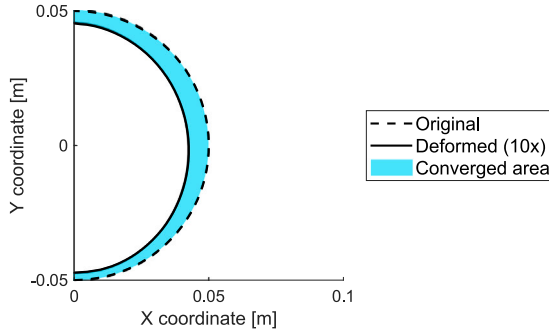


Fig. 7. Illustration of the converged area (scaled by 10).

field of plastic shear strain magnitude ($\epsilon_{s,p}$) around the cavity was used in this study as a measure of the stability of the tunnel after excavation. Following the recommendations of Atkinson and Sallfors (1991) for plasticity analysis, only plastic shear strain magnitudes above 10^{-4} were considered for the assessment. A sensitivity analysis, not presented here for brevity, was conducted to evaluate whether the extracted plastic zone was affected by further reducing the threshold (which could potentially lead to including additional portions of the plastic region surrounding the cavity). Reducing the threshold did not result in any change in the extent or morphology of the identified plastic zones. This indicates that the selected threshold adequately captures the full plastic zone, and that the material beyond this region remains elastic. The field of $\epsilon_{s,p}$ exceeding 10^{-4} can be visualized on the deformed mesh, and forms what we call the “plastic zone” in the following, as illustrated in Fig. 8. From the contours of the shear plastic strain magnitudes, we extracted the maximum plastic shear strain magnitude, noted $(\epsilon_{s,p})_{max}$, the mean of the plastic shear strain magnitude within the plastic zone, noted $(\epsilon_{s,p})_{mean}$, and the variability of the plastic shear strain magnitude over the plastic zone characterized by the coefficient of variation, noted $CV_{\epsilon_{s,p}}$. The coefficient of variation is equal to $CV = \sigma/\mu$, where μ is the mean and σ is the standard deviation. Since mean values of shear plastic strain magnitudes vary depending on the simulation run, CV was chosen over σ for the analysis. Additionally, the shape and morphology of the plastic zones were evaluated by fractal dimensions (D_b and D_{DBC}) and Minkowski functionals (M_0 , M_1 and M_2), as explained below.

Fractal dimension. A fractal is a shape, pattern, or process that exhibits self-similarity over multiple spatial or temporal scales, and provides a mathematical and quantitative description for irregular geometries and systems found in nature (Ziukelis et al., 2022). The fractal dimension (Hausdorff dimension) is defined as a ratio of the change in geometric detail to the change in scale (Cannon, 1984). Higher fractal dimensions correspond to greater geometric or pattern complexity. The

fractal dimension lumps geometric intricacies such as convexity and compactness, and is thus well-suited for describing the plastic zone induced by single-pass and staged excavation schemes.

In this study, the contour complexity of the plastic zone is quantified using the box-counting (Minkowski–Bouligand) fractal dimension. The contour is first converted to a binary image and overlaid with a grid of square boxes of various side lengths, noted ϵ . For each ϵ , the number of boxes $N(\epsilon)$ that intersect the contour is calculated. The box-counting dimension D_b , is defined as:

$$D_b = \lim_{\epsilon \rightarrow 0} \frac{\log N(\epsilon)}{\log(1/\epsilon)}. \quad (1)$$

The procedure used for calculation is illustrated in Fig. 9. Non-integer box-counting dimensions are expected to be within the interval $1 < D_b < 2$, but for a sparse set of points the dimension can fall below one (Pentland, 1984).

Measuring plastic zone contour complexity alone is insufficient for distinguishing between excavation schemes; the internal heterogeneity and variation of the plastic zone must also be quantified. To this end, we adopt the differential box-counting method (DBC) (Sarkar and Chaudhuri, 1994). Unlike binary box-counting, the DBC algorithm derives $N(\epsilon)$ from the grey-level distribution. A grey-scale image of size $M \times M$ pixels is overlaid with a square grid of box size $\epsilon \times \epsilon$ (expressed in pixels squared), where $M/2 \geq \epsilon > 1$ and $\epsilon \in \mathbb{N}$. Within each (i, j) th grid cell, the grey levels are represented by a vertical stack of $n_\epsilon(i, j)$ rectangular boxes of height h - one box per grey-level present in the cell. $n_\epsilon(i, j)$ is the number of boxes needed to cover the (i, j) th grid cell and is calculated as:

$$n_\epsilon(i, j) = \left\lfloor \frac{g_{max}}{h} \right\rfloor - \left\lfloor \frac{g_{min}}{h} \right\rfloor + 1, \quad (2)$$

where g_{max} and g_{min} denote the maximum and minimum grey levels inside the (i, j) th cell. The box height is defined as:

$$h = \frac{\epsilon G_g}{M}, \quad (3)$$

where G_g is the total number of grey levels in the image. The total box count for the image is given by:

$$N_\epsilon = \sum_{i,j} n_\epsilon(i, j). \quad (4)$$

The method to calculate N_ϵ is illustrated in Fig. 10. By plotting $\log N_\epsilon$ versus $\log(1/\epsilon)$ and estimating the slope via linear regression, we obtain the differential box-counting fractal dimension D_{DBC} . This dimension measures the grey-scale complexity and thus the internal variation of plastic zones generated by the different excavation schemes, such as compactness. Together, the binary box-counting method and the differential box-counting method provide complementary descriptors of plastic zone: the binary box-counting method characterizes the evolution and complexity of the plastic zone boundary evolution, whereas the differential box-counting method captures the internal grey-level intensity variations and heterogeneity within the plastic zone; therefore, using both yields a more complete and discriminative basis for comparing excavation schemes.

Minkowski functionals. Minkowski functionals form a set of additive, motion-invariant morphological measures that capture both the geometry and the topology of a system (Boelens and Tchelepi, 2021). They are widely used in the analysis of porous media (Armstrong et al., 2019). Minkowski functionals complement fractal metrics by describing attributes that fractal dimension alone cannot, specifically, surface area, boundary length, and connectivity. In this study, Minkowski functionals are calculated to characterize the plastic zone around the excavated cavity.

In two dimensions, there are three independent Minkowski functionals:

$$M_0(X) = \int_X ds, \quad (5)$$

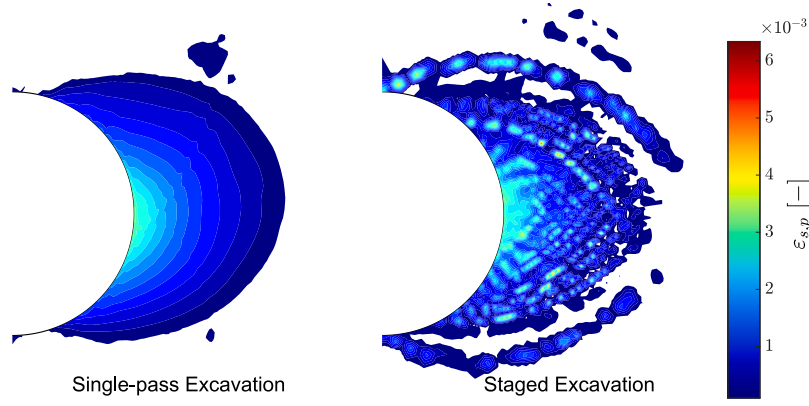


Fig. 8. Example of contours of the plastic shear strain magnitude obtained during the simulations (only plastic shear strains of magnitude above 10^{-4} are represented).

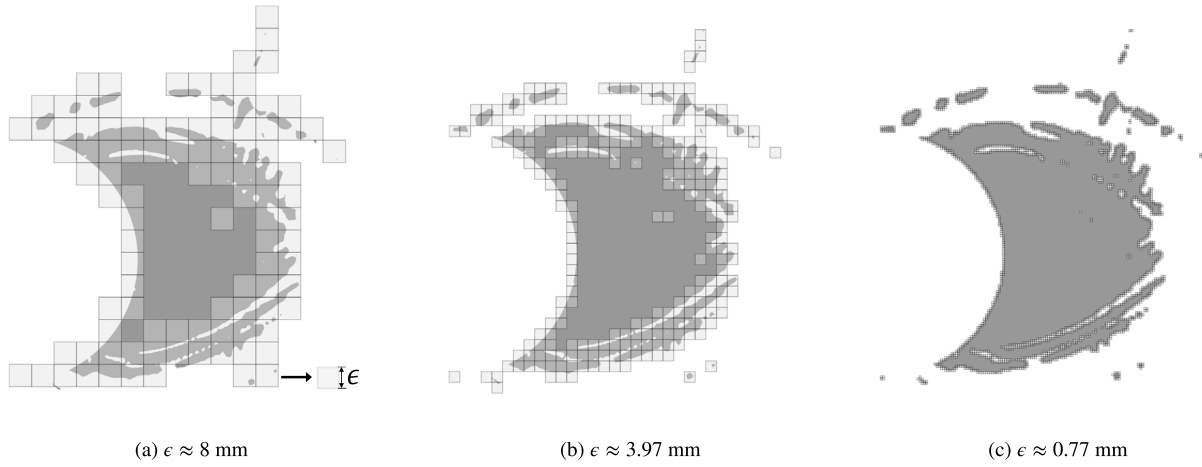


Fig. 9. Illustration of the calculation of the box-counting dimension at three scales. Here, ϵ denotes the side length of a box, and the fractal dimension is obtained by dividing $\log N(\epsilon)$ by $\log(1/\epsilon)$ for decreasing values of ϵ .

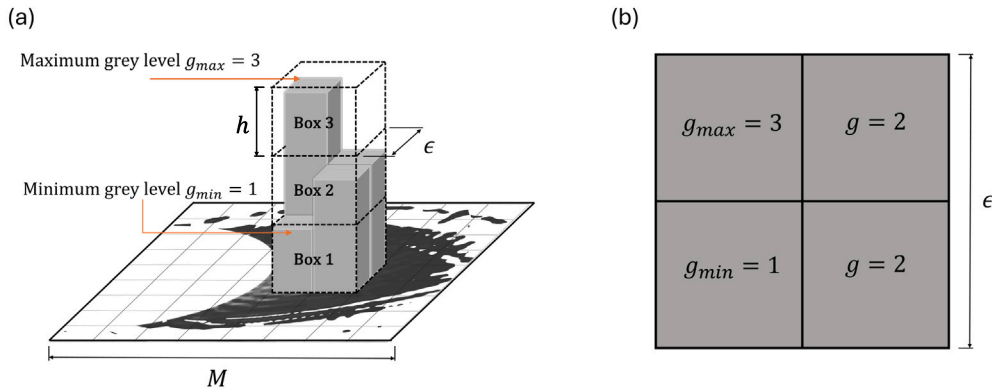


Fig. 10. Illustration of the differential box-counting (DBC) procedure. (a) The grayscale image is divided into square cells of side length ϵ in the x - y plane. For the highlighted cell, the minimum and maximum grey levels are assumed $g_{min} = 1$ and $g_{max} = 3$. Assuming a box height of $h = 1$, this cell is assigned $n_e(i, j) = g_{max} - g_{min} + 1 = 3$ boxes according to Eq. (2). (b) Top view of the (i, j) th cell showing the grey levels in each pixel. Repeating this process for all cells yields the total count $N(\epsilon) = \sum n_e(i, j)$, which is then used to estimate the fractal dimension.

$$M_1(X) = \frac{1}{2\pi} \int_{\partial X} dc, \quad (6)$$

$$M_2(X) = \frac{1}{2\pi^2} \int_{\partial X} \left[\frac{1}{R} \right] dc, \quad (7)$$

where X is the area of the plastic zone, ds is the surface element, dc is a boundary-length element along ∂X , and R is the local radius of

curvature. Eqs. (5)–(7) yield the following descriptors (Mecke, 2000): Surface area $S = M_0(X)$, Boundary length $C = 2\pi M_1(X)$, and Euler characteristic $\chi = \pi M_2(X)$. Euler characteristic, χ , is a topological invariant that summaries how a structure is connected. It can be evaluated from the first three Betti numbers, as follows:

$$\chi = \beta_0 - \beta_1 + \beta_2 \quad (8)$$

where β_0 is the number of clusters of connected zones in the plastic region, β_1 is the number of independent loops formed by these connected zones, and β_2 is the number of fully enclosed three-dimensional holes in the plastic region. For a two-dimensional planar plastic zone, $\beta_2 = 0$ and the formula reduces to $\chi = \beta_0 - \beta_1$. Together, S , C , and χ quantify the extent, roughness, and connectivity of the plastic zone induced by different excavation schemes.

3.2.2. Excavation force, torque and energy estimation

The mechanical work needed to excavate a circular tunnel is calculated from the force and torque required to advance the tunnel front by a certain distance. The calculation of the excavation force and torque is based on a combination and simplification of the analytical estimation methods proposed by Shou et al. (2010), Sheil et al. (2022), Staheli (2006) and Zhang et al. (2010). The excavation force is composed of the force needed to support the lateral earth pressure normal to the tunnel face at the excavation front (F_1) and the shear force (F_2) needed to overcome the frictional resistance around the sides of the tunnelling machine. Under the reasonable assumption that only the part of the boring machine in contact with the tunnel face is rotating, the excavation torque (T) can be calculated as the moment of tangential shear force needed to fail the material at the excavation front to enable subsequent mass removal. The excavation energy is calculated as the mechanical work done by the total thrust force ($F_T = F_1 + F_2$) and the torque. Noting δ the advancement length per unit revolution (2π), the forces, torque, and excavation energy to excavate over a tunnel length of δ are expressed in Eqs. (9) to (12)

$$F_1 = (1 - OR) \int_{A_{\text{face}}} \sigma_n dA \quad (9)$$

$$F_2 = \int_{A_{\text{side}}} \mu \sigma_R dA \quad (10)$$

$$T = CAR \int_{A_{\text{face}}} (c' + \sigma_n \tan \phi) r dA \quad (11)$$

$$W_T = \delta \underbrace{(F_1 + F_2)}_{F_T} + 2\pi T \quad (12)$$

where: OR is the opening ratio of the machine face at the excavation front; CAR is the cutting area ratio of the face describing the area fraction of the cutting bits in the excavation face; σ_n is the normal stress that is acting at the excavation front, i.e. the in-situ lateral earth pressure before excavation; μ is the coefficient of friction at the interface of the tunnelling machine sides and the soil, σ_R is the radial stress at this interface prior to loss of confinement; c' and ϕ are the apparent cohesion and friction angle of the soil, i.e., the ground strength parameters with the chosen Mohr–Coulomb plasticity model; A_{face} is the area of the face at the excavation front; A_{side} is the circumferential area, calculated as the perimeter of the boring machine multiplied by the machine length considered. If the advancement rate is too high (respectively, too low), passive (respectively, active) lateral earth pressure is generated. Therefore, the advancement rate must be adjusted to ensure that the face pressure is maintained equal to the lateral earth pressure at rest.

The advancement rate δ was assumed to be 20 cm (2D) per revolution for the purpose of this study. The side area was thus calculated for a machine length of 20 cm (for a single-pass excavation, $A_{\text{side}} = \pi \times 10 \text{ cm} \times 20 \text{ cm}$). The face area is equal to the area of the tunnel cross-section at the given excavation stage (for a single-pass excavation, $A_{\text{face}} = \pi \times 25 \text{ cm}^2$). The opening ratio and cutting area ratio were assumed to be 30% and 15% respectively, based on available information in the literature (see Godínez et al. (2015) and Italian Tunneling Society (2023)). The interface friction coefficient at the sides of the tunnelling machine is mainly a function of the soil internal friction angle, surface roughness and lubrication. It increases with the interface roughness, but cannot exceed the internal friction coefficient ($\tan \phi$) of the soil as the failure surface would move to the soil within the vicinity of the

interface, as explained in Uesugi and Kishida (1986), Staheli (2006) and Martínez (2015). A value of $0.65(\tan \phi)$ was used as the interface friction coefficient for this study.

For the single pass excavation, analytical solutions allow calculation of the normal stress σ_n on the tunnel face (of area A_{face}) and the radial stress σ_R applied on the circumference of the boring machine (over the area A_{side}) from the knowledge of the vertical stress and the lateral earth pressure at rest. However, for a staged excavation, a numerical method such as the FEM is needed to obtain the redistributed stress components σ_n and σ_R at each stage of the excavation, and thereby estimate the excavation force and torque from Eqs. (9)–(11). An example of redistributed normal stress contour used in the estimation of face force and torque is shown in Fig. 11.

The redistributed radial stresses are directly extracted from the FEM results — see Fig. 12 for example. Then, the radial stress at the cavity wall (along the perimeter of the excavation machine) is integrated over the circumferential area, A_{side} . Fig. 13 is a visualization of the radial stress extracted along the arc length of the anticipated cavity wall, before a single-pass excavation, and before each stage of the staged excavation. Fig. 13 was obtained by extracting the value of σ_R along the solid and dashed lines that represent the anticipated cavity walls in Fig. 12. The dashed line represents the anticipated cavity wall after a single-pass excavation is completed, and the corresponding radial stress along this perimeter before the excavation is shown with the dashed line in Fig. 13 labelled as “Single-pass”. The solid lines in Fig. 12 show the different positions of the cavity wall during a multi-pass excavation, and similarly, the plots in Fig. 13 show the radial stress along the anticipated cavity wall before each pass. Note that, in Fig. 13, the radial stress is plotted as function of distance along the arc, i.e., the span of the plot is larger for excavation stages for which the cavity is larger. The peaks of radial stress on either ends of each line plot are at the foot and crown of the cavity, as shown in Fig. 12. The resultant frictional force is a function of the area under the broken line for the single-pass excavation method, and the summation of areas under the solid line curves for the staged excavation method.

3.3. Statistical methods

Spearman's rank correlation coefficient. Spearman's rank correlation coefficient and Pearson's correlation coefficient quantify the relationship between variables. The Spearman rank correlation coefficient (r_s) is a non-parametric statistic that measures the strength and direction of a monotonic association between two variables. The Pearson correlation coefficient measures the linear association that exists between two continuous variables. The application of Pearson's correlation assumes that the two variables under study (X and Y) can be represented by Gaussian distributions, that the observations are independent, and that both variables are linearly related. The Pearson coefficient, noted r , is defined as:

$$r = \rho(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \cdot \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}, \quad (13)$$

where ρ is the Pearson correlation operator; x_i and y_i are the paired observations for variables X and Y , respectively; $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ are their sample means; and n is the number of paired observations. The numerator measures the covariance between X and Y . The denominator is the product of the standard deviations of the two variables X and Y . The Pearson correlation coefficient takes a value in the $[-1, 1]$ interval, where -1 indicates a perfect negative linear correlation and $+1$ indicates a perfect positive linear correlation (Berman, 2018).

Unlike the Pearson product-moment correlation, Spearman's coefficient makes no linearity assumption and instead assesses how well

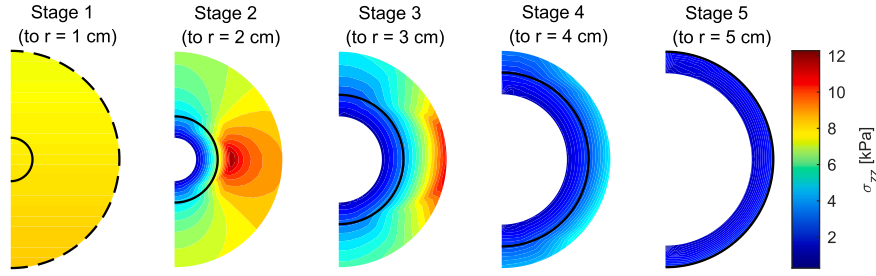


Fig. 11. Stress normal to the tunnel face ($\sigma_n = \sigma_{zz}$) encountered at the beginning of each excavation stage in the staged excavation method. Note that the distribution on the left is also the distribution of normal stress at the tunnel face at the beginning of the single-pass excavation process. The solid lines signify the outline of the excavation extent at each stage while the broken line corresponds to the outline for the single-pass method.

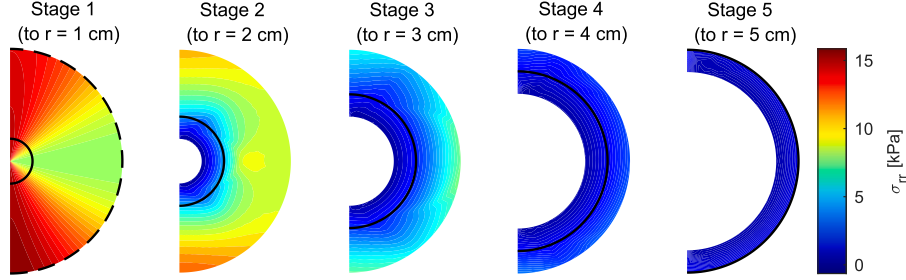


Fig. 12. Radial stress (σ_r) contour encountered at the beginning of each excavation stage in the staged excavation method. Note that the distribution on the left is also the distribution of radial stress at the tunnel face at the beginning of the single-pass excavation process. The solid lines signify the outline of the excavation extent at each stage while the broken line corresponds to the outline for the single-pass method.

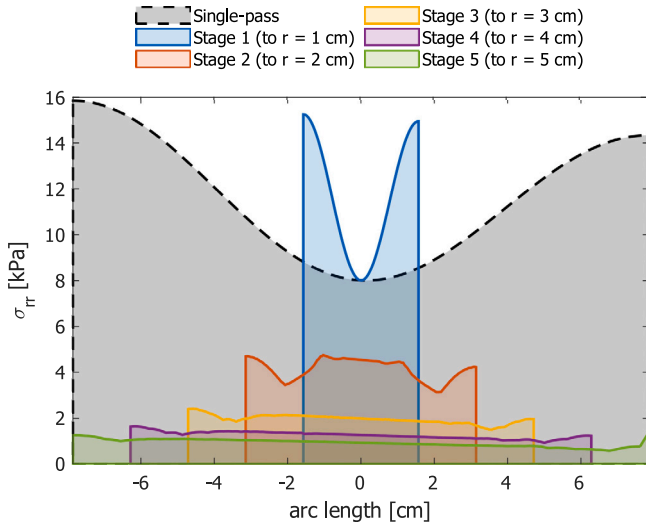


Fig. 13. Radial stress (σ_r) along the anticipated cavity wall before the single-pass excavation and before each stage of the staged excavation process. The arc length represents the perimeter of the semi-circular outline of the excavation extent.

any monotonic function can describe the relationship (Hauke and Kossowski, 2011). For a sample of size n , the paired observations $\{(x_i, y_i)\}_{i=1}^n$ are replaced by their ranks $R_{X,i}$ and $R_{Y,i}$ within X and Y . Spearman's correlation coefficient r_s is then defined as the Pearson correlation of these ranks, as follows:

$$r_s = \frac{\sum_{i=1}^n (R_{X,i} - \bar{R}_X)(R_{Y,i} - \bar{R}_Y)}{\sqrt{\sum_{i=1}^n (R_{X,i} - \bar{R}_X)^2} \sqrt{\sum_{i=1}^n (R_{Y,i} - \bar{R}_Y)^2}}. \quad (14)$$

Kernel Density Estimation (KDE) of a probability distribution function (PDF). In analysing the distributions of the output metrics — including

the plastic zone descriptors as well as the force, torque, and energy estimates — we employed the *Kernel Density Estimation* (KDE). The KDE is a non-parametric method that estimates the *Probability Density Function* (PDF) of a random variable by placing a smooth kernel at each data point and summing these contributions to obtain a continuous density curve. Compared to histograms, the KDE avoids dependence on bin edges and produces a smooth, differentiable estimate of the underlying distribution, as detailed in Silverman (1998) and Scott (2015).

For a sample of n data points $\{x_1, x_2, \dots, x_n\}$, the KDE at location x is given by:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right), \quad (15)$$

where $h > 0$ is the bandwidth (smoothing parameter) and K is the kernel function. In this study, we used a Gaussian kernel, defined in the following equation:

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}u^2\right), \quad (16)$$

which places the highest weight on points close to x and decreases smoothly according to the Gaussian bell curve. The bandwidth h governs the degree of smoothing: a small h produces a more jagged estimate sensitive to data fluctuations, while a large h can potentially over-smooth and obscure important features. Bandwidth selection was carried out using *Scott's Rule* (Scott, 2015), described by the following equation:

$$h = n^{-\frac{1}{d+4}} \sigma, \quad (17)$$

where d is the number of dimensions ($d = 1$ in this case because each data point representing an output metric is a scalar) and σ is the sample standard deviation. The KDE was calculated using the python library Seaborn in which *Scott's Rule* is also implemented as the default bandwidth selection method in the KDE function, to balance bias and variance in the estimate (see details in Scott (2015)).

All KDE analyses were performed using the `sns.kdeplot` function in the Python library Seaborn, with the following settings: `cut=0` (to avoid extrapolating beyond the data range), `gridsize=512` (for high-resolution curves), and `clip=(0, ∞)` (to restrict support to non-negative values since the negative values are outliers) as per the function options summarized in Waskom (2024). The resulting KDEs provide smooth approximations of the probability distributions for each output metric considered in this study, enabling consistent comparisons across excavation methods.

Feature importance ranking. Feature importance in this study is quantified using SHAP (SHapley Additive exPlanations) values (Lundberg and Lee, 2017). SHAP, rooted in cooperative game theory, attributes to each feature its contribution to an individual prediction. For a sample i , the model prediction admits the additive decomposition

$$\hat{y}_i = y_{\text{base}} + f(x_i, 1) + f(x_i, 2) + \dots + f(x_i, M), \quad (18)$$

where $\hat{y}_i$ is the prediction for the (i, j) th sample; y_{base} is the model's baseline output; and $f(x_i, j)$ is the SHAP value—the contribution of feature j to the prediction for sample i . Thus, each prediction is represented as the baseline plus the sum of feature-wise contributions. The predictive model is an additive ensemble of regression decision trees trained with Extreme Gradient Boosting (XGBoost). (400 boosting rounds, maximum depth 4, learning rate 0.05, row subsampling 0.8, and column subsampling 0.8). Performance is assessed with 5-fold shuffled cross-validation, feature importance is computed on each validation fold as the mean absolute SHAP value per feature and then averaged across folds to obtain the final ranking.

XGBoost is a scalable and efficient implementation of gradient-boosted decision trees (GBDT), known for its speed, regularization, and robustness (Chen and Guestrin, 2016). It constructs an additive ensemble of weak learners (e.g., decision trees) for both regression and classification. The strong learner or ensemble prediction is given by:

$$\hat{y}_i = \phi(\mathbf{x}_i) = \sum_{k=1}^K f_k(\mathbf{x}_i), \quad f_k \in \mathcal{F}, \quad (19)$$

where $\hat{y}_i$ denotes the prediction for the i th instance; $\mathbf{x}_i \in \mathbb{R}^m$ is its m -dimensional feature vector; K is the number of trees in the ensemble; $k \in \{1, \dots, K\}$ indexes the trees; and $\mathcal{F}$ denotes the function space of regression trees.

Ensemble boosting refines the residuals (i.e., the discrepancy between predictions and observations) in an iterative manner. At each iteration, the new model is built on top of the previous one by adding a tree such that the remaining error is minimized. Concretely, at iteration t , a regression tree f_t is added to greedily correct the predictions $\hat{y}_i^{(t-1)}$ from the previous stage, forming $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)$. The XGBoost objective at iteration t is

$$\tilde{\mathcal{L}}^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t), \quad \Omega(f) = \gamma_t T_t + \frac{\lambda_r}{2} \|w\|_2^2, \quad (20)$$

where n is the number of training instances; $l(\cdot, \cdot)$ is the pointwise loss; $\hat{y}_i^{(t-1)}$ denotes the prediction for the i th instance from the previous iteration; f_t is the regression tree added at iteration t which is a piecewise-constant function of the form $f_t(\mathbf{x}) = w_{q(\mathbf{x})}$, with structure q mapping $\mathbf{x}$ to a leaf index and leaf weights w ; $\Omega(f)$ is the tree-complexity regularizer that penalizes both the number of leaves and the magnitude of leaf weights. T_t is the number of leaves in f_t ; $w = (w_1, \dots, w_T)$ is the vector of leaf weights; $\gamma_t \geq 0$ penalizes the number of leaves; $\lambda_r \geq 0$ is the L_2 regularization coefficient on leaf weights; and $\|w\|_2^2 = \sum_{j=1}^{T_t} w_j^2$.

Applying a second-order Taylor expansion of the pointwise loss around the previous predictions $\hat{y}_i^{(t-1)}$ and discarding terms that are constant with respect to f_t yields:

$$\tilde{\mathcal{L}}^{(t)} = \sum_{i=1}^n \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \Omega(f_t), \quad (21)$$

where $g_i = \partial_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)})$ and $h_i = \partial_{\hat{y}^{(t-1)}}^2 l(y_i, \hat{y}^{(t-1)})$ are the first- and second-order derivatives of the pointwise loss.

Let $f_t(\mathbf{x}) = w_{q(\mathbf{x})}$ be the candidate tree at iteration t , with structure $q: \mathbb{R}^m \rightarrow \{1, \dots, T\}$ and leaf weights $w = (w_1, \dots, w_T)$. Let us define the instance index set of leaf j as $I_j = \{i: q(\mathbf{x}_i) = j\}$ for $j = 1, \dots, T$. Then Eq. (21) can be rewritten as

$$\tilde{\mathcal{L}}^{(t)} = \sum_{j=1}^{T_t} \left[\left(\sum_{i \in I_j} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda_r \right) w_j^2 \right] + \gamma_t T_t. \quad (22)$$

Fixing the tree structure $q(\cdot)$, the optimal weight for leaf j is

$$w_j^* = - \frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda_r}, \quad j = 1, \dots, T_t, \quad (23)$$

Substituting w_j^* into the second-order objective yields the optimal value for the given structure at iteration t :

$$\tilde{\mathcal{L}}^{(t)}(q) = - \frac{1}{2} \sum_{j=1}^{T_t} \frac{\left(\sum_{i \in I_j} g_i \right)^2}{\sum_{i \in I_j} h_i + \lambda_r} + \gamma_t T_t. \quad (24)$$

4. Results and discussion

4.1. Spearman's rank correlation coefficients

Fig. 14 displays the Spearman's correlations between geotechnical parameters (depth to diameter ratio and soil constitutive properties), geometric characteristics of the plastic zone, and estimates of the mechanical work needed for excavation. Fig. 14(a) shows the correlations in the single-pass excavation method and Fig. 14(b) shows the correlations in the staged excavation method while Fig. 14(c) shows the difference in correlations demonstrating the change of correlations as a result of changing the excavation method. The deeper shades of green (higher positive magnitudes) show higher positive correlation in Figs. 14(a) and 14(b) while in Fig. 14(c), it signifies an increase in correlation as a result of switching from the single-pass excavation method to the staged excavation method. On the other hand, the deeper shades of violet (higher negative magnitudes) show higher negative correlation in Figs. 14(a) and 14(b) while in Fig. 14(c), it signifies a reduction in correlation as a result of switching from the single-pass excavation method to the staged excavation method.

The depth-to-diameter ratio (D/d) is a physical parameter that significantly affects the plastic zone and the mechanical work needed for excavation. In both excavation schemes (Figs. 14(a) and 14(b)), D/d exhibits a strong positive monotonic correlation with the peak, mean and coefficient of variation of the plastic shear strain as well as the total thrust required to overcome side friction and face pressure. Notably, the increase in total energy, torque and force with increasing D/d is lower for a staged excavation as shown in Fig. 14(c). In other words, the excavation process used by ants, in which the tunnel cross-section is enlarged in several passes, becomes more energy-efficient for microtunnels with large depth-to-diameter ratio.

The soil bulk unit weight (γ) shows no systematic monotonic trend with the key plasticity variables in either scheme as shown in Figs. 14(a) and 14(b). The sensitivity of the total excavation energy, torque and thrust to γ is lower for a staged excavation than for a single-pass mass removal as shown in Fig. 14(c). This observation indicates that the staged excavation process is slightly more advantageous than the single-pass one, even in heavier ground. The weak correlation between the unit weight and each output evaluation metrics is likely due to the small range of magnitudes selected for the unit weight of the soil, which was constrained to represent loose unsaturated sand.

For both mass removal processes, the elastic modulus E correlates negatively with the peak, mean and CV of the plastic shear strain magnitudes ($(\epsilon_{s,p})_{\text{max}}$, $(\epsilon_{s,p})_{\text{mean}}$, and $CV_{\epsilon_{s,p}}$, respectively), which indicates that stiffer ground constrains both the magnitude and spatial variability

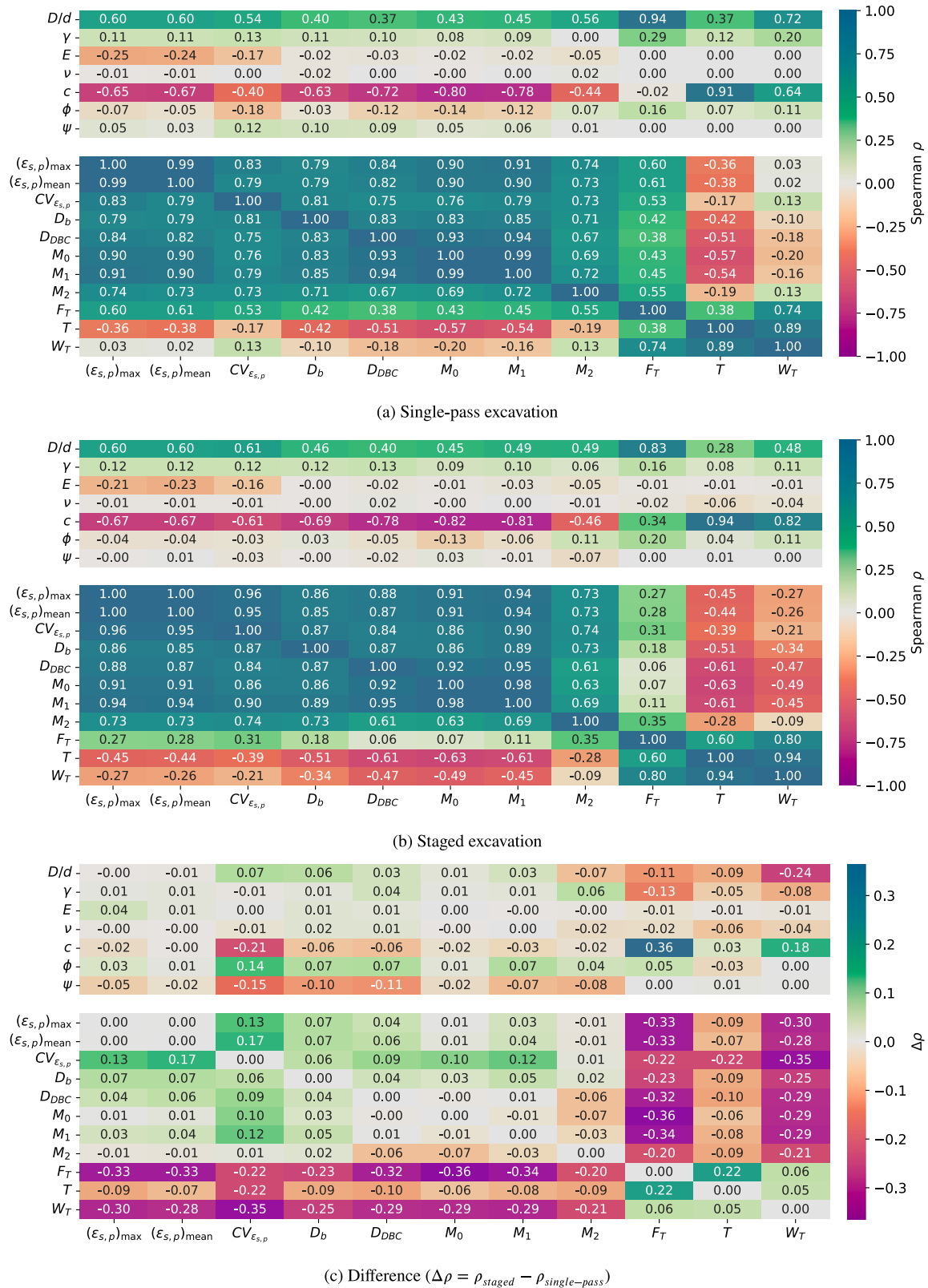


Fig. 14. Comparison of Spearman correlation coefficients for both excavation methods and their difference. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

of plastic deformation. However, neither E nor ν correlates significantly to the mechanical excavation parameters.

Figs. 14(a) and 14(b) show that apparent cohesion (c'), an important measure of inherent shear strength, is strongly negatively

correlated with each plastic strain indicator, fractal dimension, and Minkowski functional, which means that a higher apparent cohesion produces simpler, less variant plastic zones. Conversely, c' shows a strong positive monotonic correlation with the total thrust and

work, reflecting the greater effort required to excavate more cohesive medium. Fig. 14(c) further demonstrates that, for the same increase in apparent cohesion, transitioning from a single-pass to a staged excavation method leads to a stronger negative correlation between c' and the coefficient of variation (CV) of plastic shear strain, which characterizes the variability of the plastic zone. In practical terms, for cases in which plasticity is induced under a lower apparent cohesion, staged excavation results in a higher CV than single-pass excavation. This behaviour arises because staged excavation promotes progressive stress redistribution and changes in the loading paths at the material points around the cavity, leading to the formation of concentrated plastic shear strains along shear bands. These localized strain concentrations increase the variability of shear strain magnitudes within the plastic zone. In contrast, single-pass excavation induces stress redistribution and plasticity in a single stage, resulting in a more uniformly distributed plastic shear strain field with lower variability. As a result, although apparent cohesion and plastic shear strain variability are negatively correlated for both excavation methods, this negative correlation is more pronounced for staged excavation due to increased strain localization. Similarly, a stronger positive correlation between the apparent cohesion and the total excavation force—therefore energy—exists in the staged excavation method. Both of those observations are likely due to the highly variable shear strains and important stress redistributions that occur during staged excavation in soil of low apparent cohesion.

In both methods, the friction angle (ϕ) correlates positively with the total thrust, torque and energy, but only weakly. The weak negative correlation of ϕ with the coefficient of variation of plastic shear strain diminishes in the staged excavation while for ψ , it is the weak positive correlation that diminishes as shown in Fig. 14(c). For the staged excavation simulations, neither ϕ nor ψ retains control over the geometric descriptors of the plastic zone.

To summarize, both excavation schemes are highly influenced by the depth-to-diameter ratio and the apparent cohesion parameters, which control the stresses around the cavity and the shear strength of the soil, respectively. For the single-pass excavation scheme, increases in D/d or overburden increase radial and hoop stresses around the cavity and induce a significant increase in thrust and energy demand. During a staged excavation, stress redistributes each time mass is removed. This progressive and successive unloading contributes to the formation and widening of a self-supporting arching structure that transfers a portion of the overburden away from the cavity wall, which dramatically lowers the radial stress and lateral earth pressures around the cavity wall. This reduces the torque, thrust, and energy required per unit length of tunnel excavated in most situations, while yielding a more fragmented plastic zone shape.

Because arching effects only hold when the surrounding ground is not failing in shear, the staged excavation process exhibits greater sensitivity to the shear failure parameter, i.e., apparent cohesion c' . The excavation energy demand associated with staged excavation increases significantly with c' ; at sufficiently high c' , the initiation and development of arching are hindered, diminishing the energy efficiency of staging. Overall, the staged excavation method can be more energy efficient than the single-pass one for specific geotechnical conditions.

4.2. Effect of excavation method on post-excavation state

Simulation results indicate that the fraction of the tunnel cross-section area that has converged (i.e., displaced inward) is 2% or less in all the cases under study. Tunnel stability can be assessed using tunnel strain, defined as the ratio of radial tunnel wall displacement u_R to tunnel radius r , $\epsilon = u_R/r$, a deformation-based measure originally proposed by Sakurai (1981). Field measurements from case histories indicate that tunnel strains exceeding approximately 1% are associated with an increased risk of instability, as noted by Hoek (2001). For a circular tunnel, this corresponds to an area reduction of approximately 2%, since $\left(\frac{\Delta A}{A} = \frac{\pi r^2 - \pi(r-u_R)^2}{\pi r^2} \approx \frac{2u_R}{r}\right)$. Therefore, convergence was discarded from the analysis, due to its low magnitude that is not able to describe stability in a meaningful way.

4.2.1. Effect of the excavation method on the plastic zone

Maximum, mean and coefficient of variation of plastic shear strain $((\epsilon_{s,p})_{max}, (\epsilon_{s,p})_{max}$ and $(CV)_{\epsilon_{s,p}}$). The maximum and mean values as well as statistical and geometric descriptors of the plastic zone are calculated in the final stage of all the simulations run in this study. The KDEs for the maximum, mean, and coefficient of variation (CV) of plastic shear strain are shown in Fig. 15. For both the mean and maximum plastic shear strain, the single-pass method exhibits lower peak values and smaller magnitudes in most simulation cases, whereas the staged method shows higher values and occasional extreme outliers. The CV distributions are generally similar for both methods, but the staged method displays greater extremes, indicating higher spatial variability of plastic deformation, likely associated with localized shear bands. These trends are consistent with the observations made in Section 4.1, where it was noted that the staged method produces more fragmented and heterogeneous plastic zones. The distributions of maximum plastic shear strain show spikes at zero for both cases, indicating that the majority of cases — particularly for the single-pass method — exhibit little to no plasticity. The mean plastic shear strain values are similarly concentrated near zero, reflecting the generally small average plastic shear magnitudes around the cavity. This suggests that, although plastic shear strains occur, their magnitude is minor in most cases and unlikely to be of significance for engineering design purposes. The plastic shear strain CV distributions for both methods peak near 0.8, indicating comparable variability in plastic shear strain magnitudes within the plastic zone. For all three descriptors, the value of the descriptor at which the peak is reached is the same for both excavation methods, and the distributions almost match outside of the peaks particularly for the coefficient of variation. This indicates that the observed differences between the excavation methods are limited to a subset of cases rather than being a general trend.

Box counting and differential box counting fractal dimensions (D_b and D_{DBC}). For both excavation schemes, the probability density function (PDF) of the box-counting fractal dimension (D_b) of the plastic zone exhibits a narrow, dominant peak centred at $D_b \approx 1.05$ (Fig. 16(a)), indicating that the plastic zone boundary is nearly Euclidean. A secondary peak near $D_b = 0$ corresponds to simulations in which no plasticity develops. At the right end of the distribution, the PDF for the staged excavation method extends beyond that of the single-pass method, up to $D_b \approx 1.30$. In summary, the gross silhouette of the plastic zone differs little between the two excavation processes; only a small subset of staged excavation simulations yield marginally richer edge detail.

Fig. 16(b) shows that the PDF of the differential box-counting dimension (D_{DBC}) differs between the two excavation schemes. The staged excavation PDF is broader than that of the single-pass method, with its upper quartile extending beyond the single-pass distribution and reaching up to $D_{DBC} \approx 2.4$. This indicates that staged excavation tends to produce plastic zones with greater and more variable internal complexity. Both methods, however, exhibit two common peaks near $D_{DBC} \approx 2.0$ and $D_{DBC} \approx 2.1$. A value of $D_{DBC} \approx 2.0$ corresponds to cases where the plastic shear strain map is blank (a perfectly flat intensity surface), indicating no plasticity; these cases also correspond to $D_b = 0$ in the box-counting analysis. The second peak (≈ 2.1) corresponds to cases where plasticity occurs but with relatively regular morphology. The higher-end values observed for staged excavation represent a small subset of simulations in which the plastic zones are notably complex and fragmented.

In summary, the two excavation schemes yield plastic zones that have similar contour shapes, which are essentially Euclidean, for a majority of the cases. But the plastic zone is more complex for the staged excavation: it is typically not compact, and presents a hierarchy of finer, intersecting shear bands, raising the differential fractal dimension and occasionally producing highly intricate patterns. Noting that D_{DBC} is calculated in a similar way to D_b but with more information encoded from the grey-level variability within the plastic zone, the difference between the two methods is more visible in D_{DBC} . Therefore, D_b will be eliminated from the subsequent analysis.

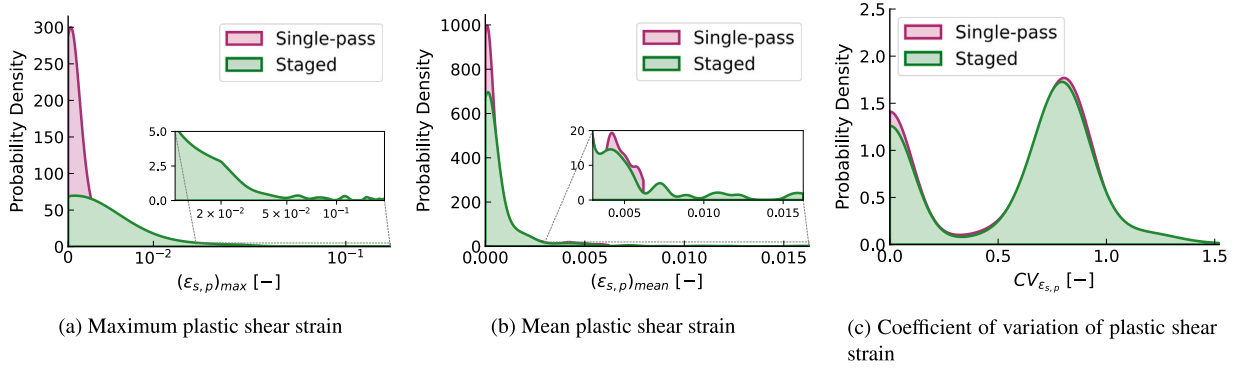


Fig. 15. Maximum and mean values of plastic shear strain magnitude and coefficient of variation of the plastic shear strain for both excavation methods.

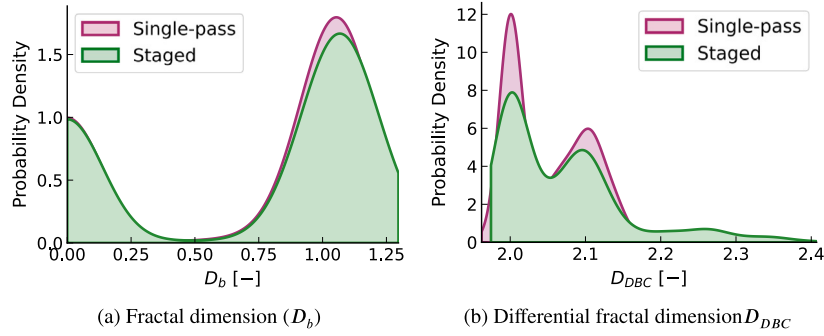


Fig. 16. Comparison of two fractal analysis methods for quantifying the complexity of the plastic zone geometry for both excavation methods.

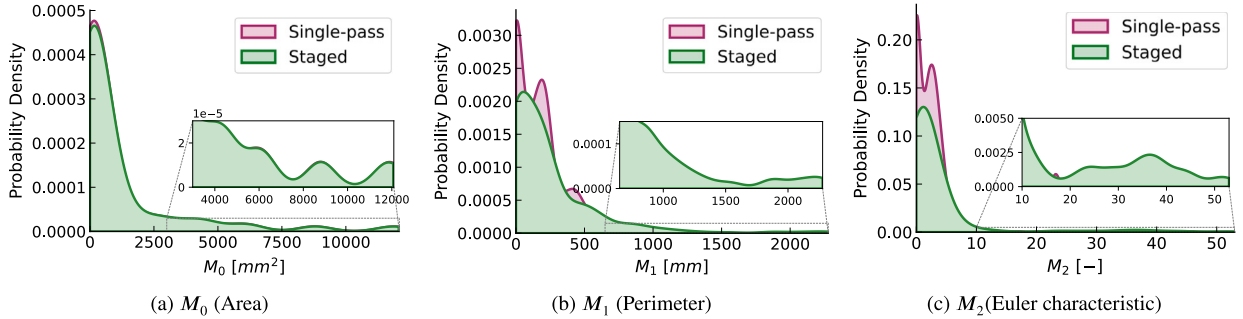


Fig. 17. Comparison of the morphology and topology of plastic zones induced by both excavation methods using Minkowski functionals.

Minkowski functionals: M_0 (Area in mm^2), M_1 (perimeter in mm), and M_2 (Euler characteristic). The difference in topology and complexity of plastic zones is also reflected by the Minkowski functionals PDFs, presented in Fig. 17. The area M_0 exhibits virtually identical distributions for both excavation schemes and will not be included in the subsequent analysis as it does not provide information on the effect of the excavation method. The perimeter M_1 shows a broader distribution for the staged excavation process, with a pronounced long tail towards larger values. This indicates that staged excavation more frequently produces plastic zones with highly tortuous boundaries. The Euler characteristic M_2 also exhibits a long tail towards large values in the staged excavation method, indicative of fragmented plastic zones. In the staged excavation method, the complex plastic zones are characteristic of localized strain energy dissipation and stress redistribution over several successive excavation steps, which may facilitate the excavation process. This observation is congruent with current studies of ground reaction behaviour that show that, as the radius of the plastic zone (R_p) increases, the ground-reaction curve flattens and the self-support capacity of the surrounding ground significantly decreases (see Ramoni and Anagnostou (2011) and Zheng et al. (2024)).

4.2.2. Ranking of geotechnical features by importance to the plastic zone

Fig. 18 is obtained from the feature selection algorithm described in and it illustrates the relative importance of the input parameters in determining the magnitude and variability of the plastic shear strains. In agreement with the trends observed from the Spearman correlation analysis, apparent cohesion (c'), depth-to-diameter ratio (D/d), and elastic modulus (E) emerge as the most influential parameters across both excavation schemes. Among these, apparent cohesion and depth-to-diameter ratio remain the primary controlling factors for the coefficient of variation of plastic shear strain—in both methods. This is because higher apparent cohesion increases the soil resistance to yielding, thereby reducing the extent and variability of plastic zones, while the D/d ratio influences the stress level around the cavity, resulting in higher magnitudes and variability of plastic shear strain with increasing D/d . The elastic modulus is inversely related to the plastic shear strain magnitudes: stiffer ground experiences smaller stress redistribution and reduced deformation, resulting in smaller plastic shear strain magnitudes.

An interesting observation from the parameter sensitivity analysis of maximum plastic shear strain is that the internal friction angle and

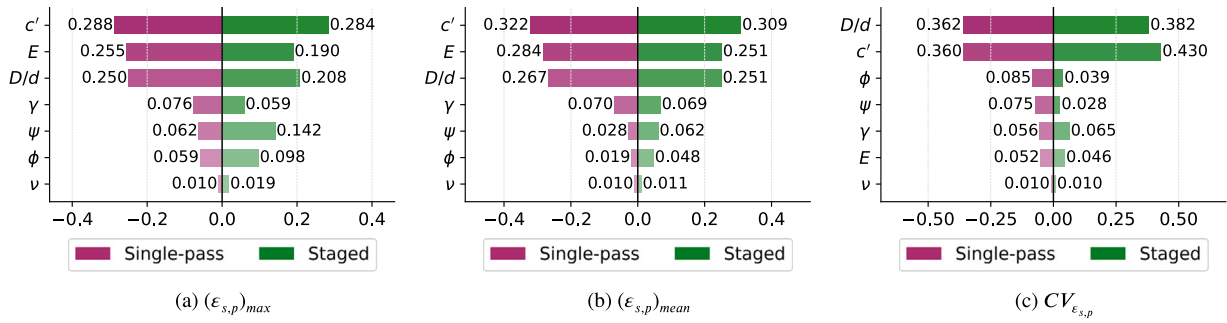


Fig. 18. Feature importance ranking for $(\epsilon_{s,p})_{max}$, $(\epsilon_{s,p})_{mean}$, and $CV_{\epsilon_{s,p}}$.

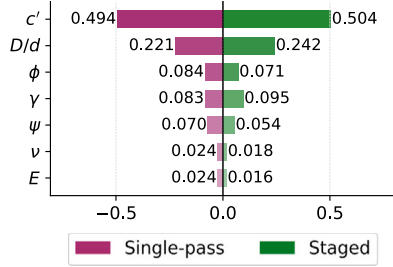


Fig. 19. Feature importance ranking for differential box-counting fractal dimension (D_{DBC}).

dilation angle become more influential when shifting from a single-pass to a staged excavation scheme as shown in Fig. 18(a). This increased importance is likely due to the multiple phases of stress redistribution in staged excavation, where each elementary volume in the surrounding ground is unloaded incrementally. Consequently, soil material points begin each stage from different stress states and follow distinct stress paths before yielding. In the Mohr–Coulomb model, the friction angle governs the shear strength mobilized along each stress path, while the dilation angle controls the associated volumetric plastic strain. Therefore, changes in the starting stress state alter the effect of the friction and dilation angles during subsequent unloading. The shear strains and volumetric changes associated with these repeated unloading events are cumulative, amplifying the influence of friction and dilation angles on the maximum shear strain. This staged unloading also contributes to the more variable and localized shear strain patterns, including the formation of shear bands around the cavity, as shown in Fig. 8.

Fig. 19 shows that the complexity of the plastic zones, measured by the fractal dimension D_{DBC} , is most influenced by the apparent cohesion and depth-to-diameter ratio. This shows that for each excavation method, the complexity of the plastic zone geometry increases with the extent of plastic zone, which itself increases with larger depth (i.e., larger stress) and lower apparent cohesion (i.e., lower soil strength).

Fig. 20 shows the ranks of the input parameters based on their effect on Minkowski functionals. The depth-to-diameter ratio, apparent cohesion, and internal friction angle control the Minkowski parameters M_1 and M_2 . This was expected, since M_1 measures the perimeter and M_2 measures the complexity of the plastic zone geometry, and both increase with increasing plastic zone size, i.e. higher stresses and lower shear strength.

It is noteworthy that the unit weight of the soil (γ) and Poisson's ratio (ν) do not emerge as influential parameters for any of the output metrics describing the plastic zones. This is likely attributable to the relatively narrow ranges of values adopted for these parameters in this study, which focused on the soils where Harvester ants burrow their nests. The relatively small interval chosen for γ and ν may limit the observable impact of these parameters on the simulated responses.

4.2.3. Effect of soil type on the plastic zones obtained with the two excavation methods

The different output metrics that describe the plastic zone are presented in Figs. 21 to 26 in one-to-one plots (agreement between single-pass and staged excavation), for the most influential parameters (c' , D/d , ϕ and ψ) identified in Section 4.2.2. Each point in these plots is a scalar metric that represents a simulation case and where it lies relative to the equivalence line shows the difference between the two methods. All points above the equivalence line are cases where the metric describing plastic shear strain around the cavity is of higher magnitude when the method of excavation is changed from single-pass to staged whilst keeping all input parameters the same. The reverse is true for points that lie below the equivalence line while the cases that lie on the equivalence line are cases where the change in excavation method has no effect of the output metric being considered. The points are encoded by different markers and colours based on the input parameters that have been deemed most influential in controlling the effect of changing the excavation method. This helps identify what kinds of soils are sensitive to the excavation method. Plots that include all the input and output parameters tested in this study can be found in the supplementary materials.

A comparison of the maximum plastic shear strains obtained from each simulation is presented in Fig. 21. From the scatter plots grouped by influential input parameters, we observe that, in the staged excavation simulations, the largest peak plastic shear strains arise for a combination of low apparent cohesion (1 kPa), deep embedment ($D/d = 6$ and 10), high friction angle ($\phi = 33^\circ$ and 38°), and very low dilation angle $\psi = 0.05\phi$. This combination promotes shear-dominated plastic mechanisms at depth and prevents a uniform distribution of plastic strains, such that plasticity concentrates into a narrow band. Plastic shear strains reach a peak at each stage of excavation which accumulates to a larger magnitude than in the corresponding cases with the single-pass excavation method. The average shear strain magnitudes (Fig. 22) follow a similar trend as the peak shear strain magnitudes (but with smaller deviation from the equivalence line, since the shear strain magnitude is averaged over the entire plastic zone). The occurrence of strain localization and shear band formation is further supported by the coefficient of variation comparisons in Fig. 23, which highlights the increased variability of the plastic shear strain values within the plastic zone for the staged excavation method.

Figs. 24 to 26 describe the fragmentation, perimeter, and complexity of the plastic zone geometry. The deviation of these metrics from the equivalence line and towards the upper half of the one-to-one agreement plots is consistent with the observation of localized shear bands forming due to the combination of parameters that promote shear-dominated plastic mechanisms that hinder the formation of a uniform plastic zone.

Table 2 reports the ratios of plastic shear strain measures obtained with a staged excavation by the corresponding plastic shear strain measure obtained with a single-pass excavation. Only simulation cases for which the plastic zone descriptors are different in the two excavation

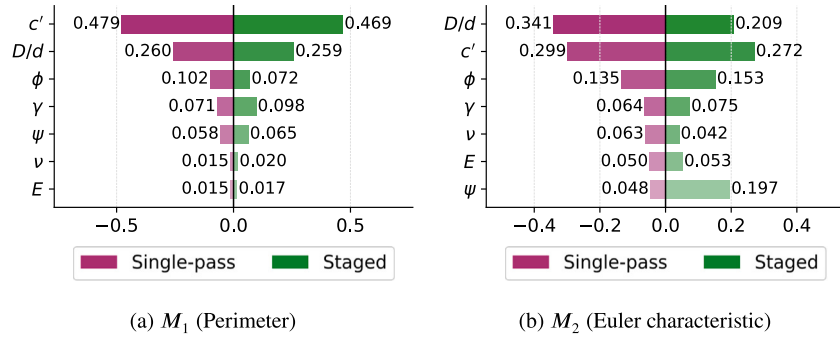


Fig. 20. Feature importance ranking for Minkowski functionals.

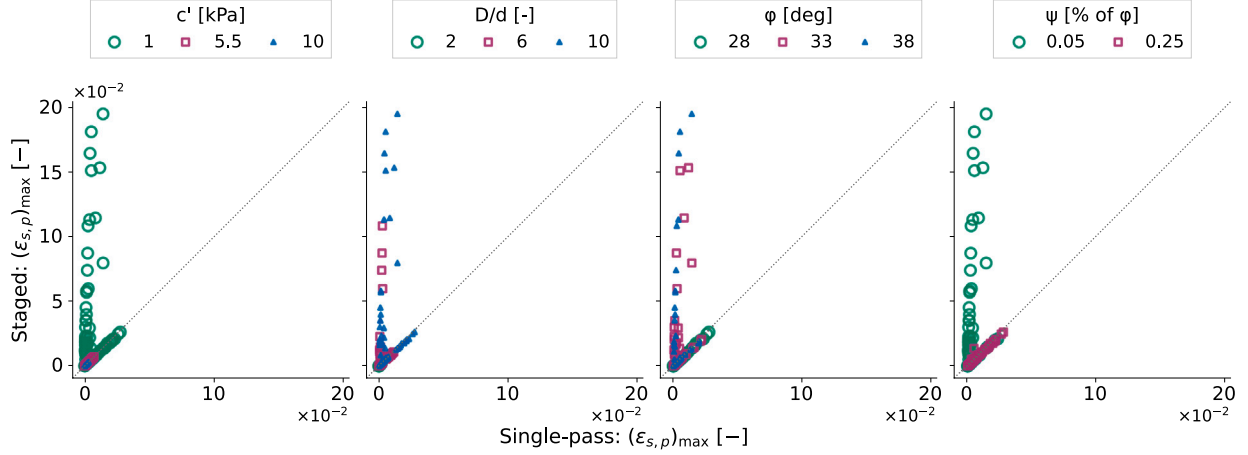


Fig. 21. Values of $(\epsilon_{s,p})_{\max}$ obtained with both excavation methods, grouped by influential model parameters. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

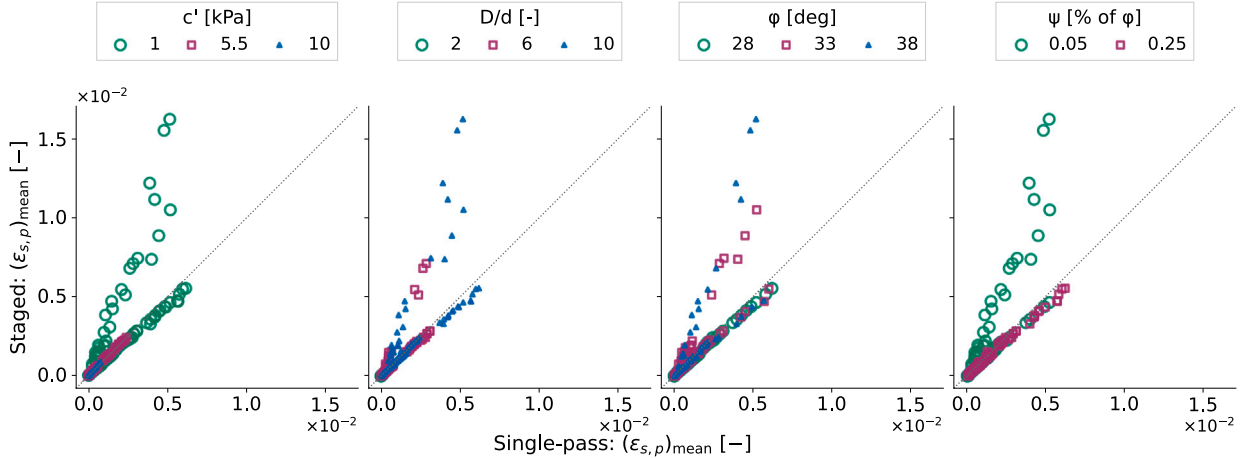


Fig. 22. Values of $(\epsilon_{s,p})_{\text{mean}}$ obtained with both excavation methods, grouped by influential model parameters. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

methods (i.e., points of the scatter plots away from the equivalence line in Figs. 21–26) are considered. Specifically, only cases for which $c' = 1$ kPa $D/d = 6$ or 10 , $\phi = 33^\circ$ or 38° $\psi = 0.05\phi$ are shown in Table 2. For these cases, the plastic measures are either higher in the staged excavation method, or nearly unchanged. The peak plastic shear strain is increased by a maximum of 40 times for intermediate depth-to-diameter ratios while for higher values of D/d , it is increased by 5 to 42 times. In average, the increase is 10 to 30 times over both depth-to-diameter ratios. A similar pattern is observed for the mean plastic

shear strains. The fractal dimension D_b increases by a factor of 1 to 2, and D_{DBC} increases by 1 to 1.2 when switching from the single-pass to the staged excavation method. The plastic zone perimeter is increased by a maximum of 8 times for intermediate depth-to-diameter ratios while for higher values, it is increased by 5 to 92 times. On average, the increase is 3 to 13 times over all realizations with both depth-to-diameter ratios. Overall, plastic shear strain magnitudes and geometry descriptors adopted in this study are generally higher in the staged excavation simulations than in the single-pass excavation simulations.

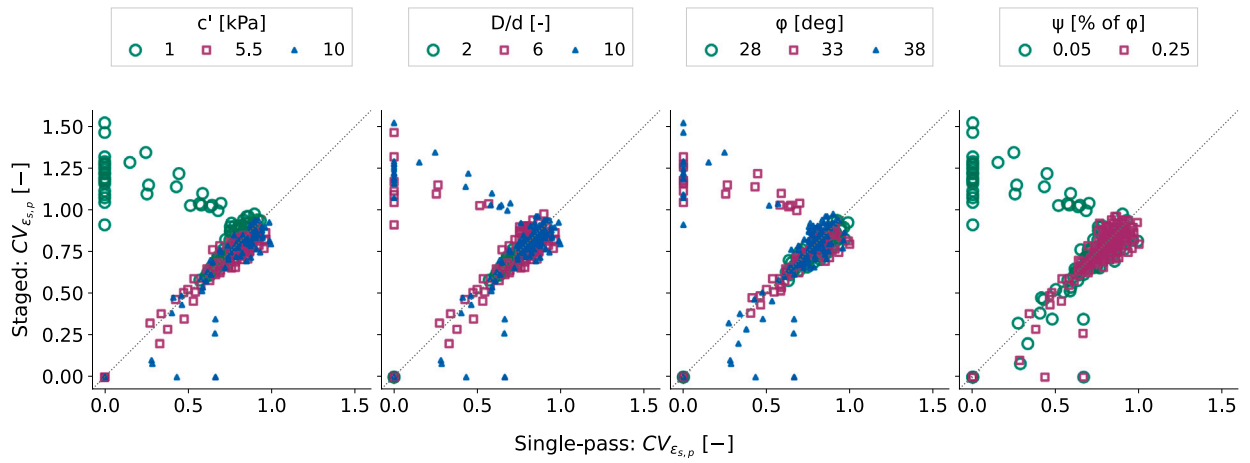


Fig. 23. Values of $(CV)_{\epsilon_{s,p}}$ obtained with both excavation methods, grouped by influential model parameters. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

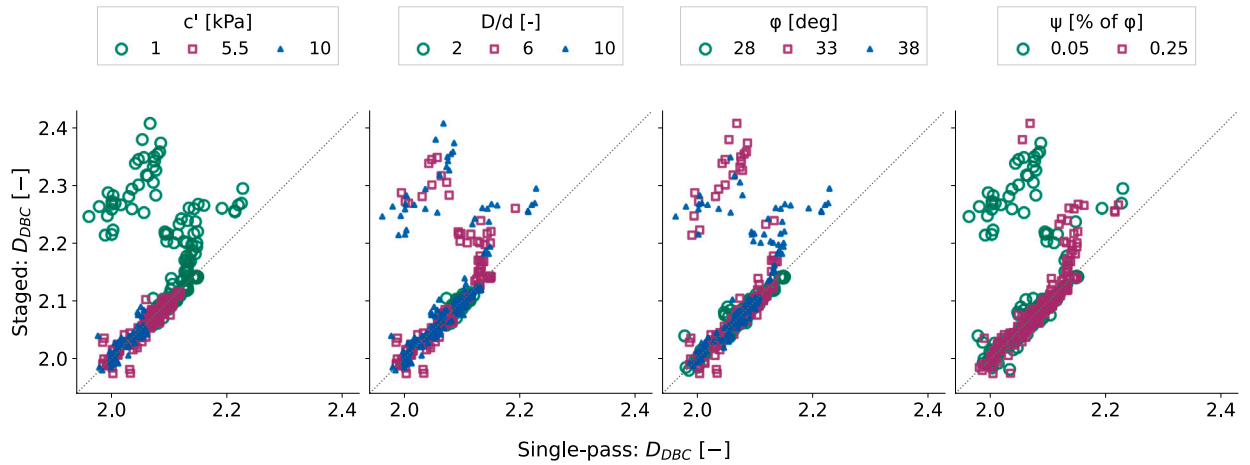


Fig. 24. Values of D_{DBC} obtained with both excavation methods, grouped by influential model parameters. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

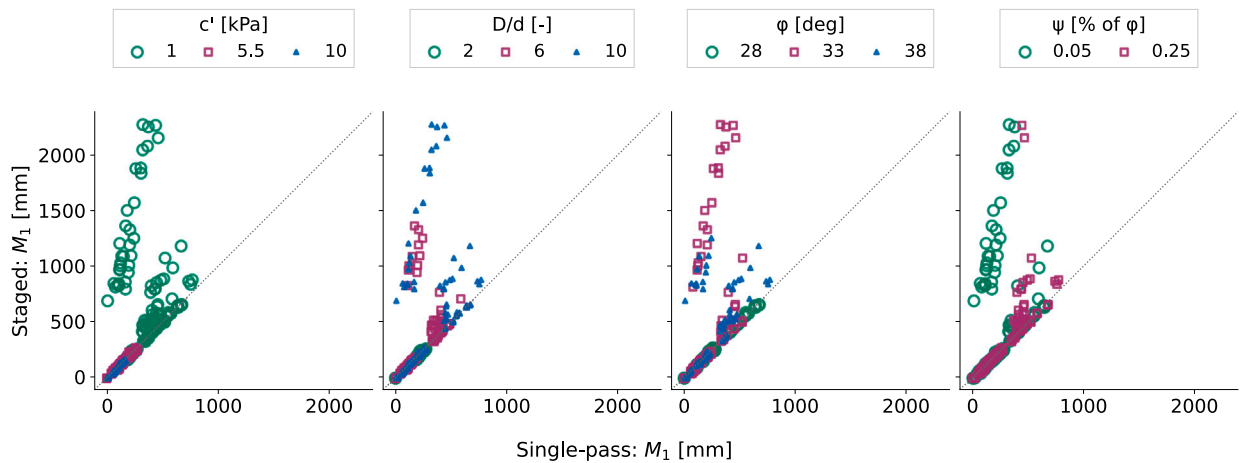


Fig. 25. Values of M_1 (plastic zone perimeter) obtained with both excavation methods, grouped by influential model parameters. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

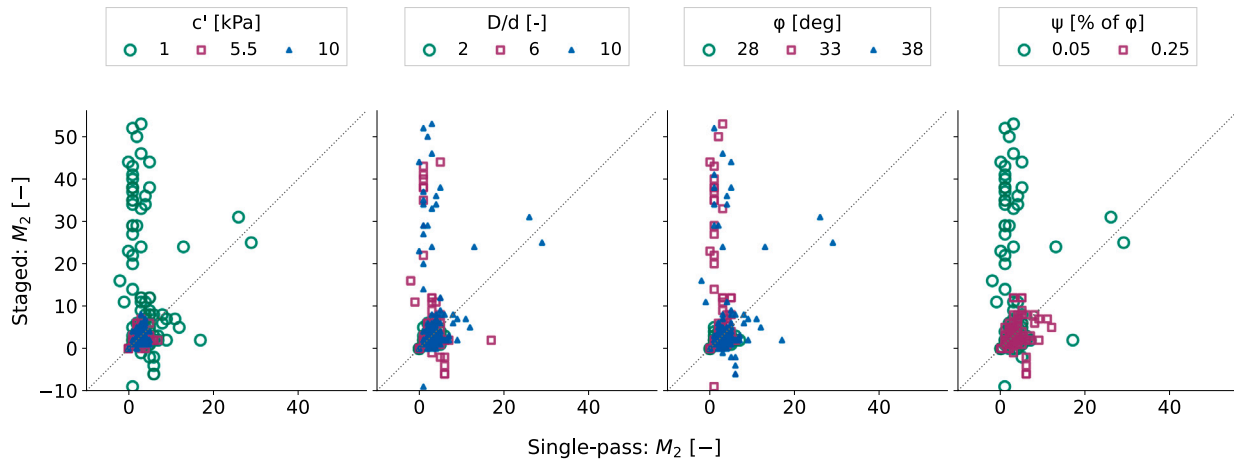


Fig. 26. Values of M_2 (plastic zone Euler characteristic) obtained with both excavation methods, grouped by influential model parameters. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

Ratios of selected plastic zone measures for simulation cases with input parameters $c' = 1$ kPa and $\psi = 0.05\phi$, with varying D/d (6 or 10) and ϕ (33° or 38°).

D/d	ϕ	Value	$(\epsilon_{s,p})_{max} : \frac{\text{staged}}{\text{single-pass}}$	$(\epsilon_{s,p})_{mean} : \frac{\text{staged}}{\text{single-pass}}$	$D_b : \frac{\text{staged}}{\text{single-pass}}$	$D_{DBC} : \frac{\text{staged}}{\text{single-pass}}$	$M_1 : \frac{\text{staged}}{\text{single-pass}}$
6	33°	Min	1.000	0.930	0.998	0.999	0.997
		Max	36.848	3.139	1.270	1.146	8.398
		Mean	12.368	1.744	1.099	1.072	4.292
	38°	Min	1.048	0.937	1.034	1.031	1.212
		Max	40.849	2.644	1.228	1.142	7.613
		Mean	10.485	1.609	1.113	1.079	3.108
10	33°	Min	5.560	1.792	1.128	1.109	5.682
		Max	42.407	2.537	1.294	1.138	10.515
		Mean	16.820	2.039	1.242	1.125	7.342
	38°	Min	12.274	2.062	1.039	1.019	1.667
		Max	40.640	3.479	2.027	1.145	92.194
		Mean	30.612	2.830	1.197	1.101	13.442

4.3. Effect of excavation method on required excavation energy

4.3.1. Effect of the excavation method on the mechanical excavation descriptors

The jacking force is calculated as the sum of the force needed to support the lateral earth pressure at the excavation front and the force needed to overcome the frictional force around the tunnelling machine, while the excavation torque is calculated as the torque needed to shear the soil to failure at the tunnel face. In a staged excavation, successive stress redistributions generally reduce the normal stress at the excavation front, lowering the torque needed to shear the material and the force needed to support that normal stress (lateral earth pressure). However, the surface over which side friction occurs increases as the diameter of the tunnel face increases during the staged excavation process. Explicit calculations are thus needed to assess the potential benefit of staged excavation in reducing the frictional component of the jacking force, and in reducing the total excavation energy needed to build the tunnel. The total excavation energy is calculated as the sum of the work done by the force during an advancement δ per revolution and the work that the torque does over this revolution.

Fig. 27(a) shows that there are more staged excavation simulation cases than single-pass excavation cases for which the total jacking force is lower than 200 N, which indicates that staging excavation may save energy. However, the simulation cases in which the total jacking force is the highest are for staged excavation, which indicates that staging excavation may also increase the total jacking force needed for excavation due to the cumulative area over which interface frictional resistance is integrated. In those simulations, the side friction component accounts for the majority of the total jacking force magnitude.

The stress redistribution lowers both the lateral earth pressure at the face and the radial stress at the side of the tunnelling machine. For this reason, the force needed to support the lateral earth pressure is always reduced when switching from single-pass to staged excavation, because the total area over which the face pressure is integrated is the same for both excavation methods. There is a small number of cases where the redistribution of stresses induced by staged excavation leads to a reduction of the radial stress around the tunnel wall but does not compensate for the increased area of frictional resistance: the total frictional force increases when switching to the staged method because the frictional component of the jacking force in a staged excavation process is integrated over a larger area than in the single-pass excavation method. In those cases, a larger jacking force is needed to excavate in stages than with a single pass.

Fig. 27(b) shows that the torque needed to fail the soil at the tunnel face with the staged excavation method never exceeds that in the single-pass excavation method, which was expected, since the total area of the surface over which the torque is applied is the same in both excavation methods. The torque probability density functions exhibit two peaks with both excavation methods. The peaks in the staged excavation method are shifted towards lower torque values compared to those in the single-pass excavation method, which suggests that excavation in stages requires less torque than single-pass excavation in some cases.

The total energy demand is calculated as the work done by both the jacking force and the face shearing torque. The PDFs of the total energy demand calculated for both excavation methods are shown in Fig. 27(c). The energy value at the peak for the staged excavation method is about half of that for the single-pass excavation method,

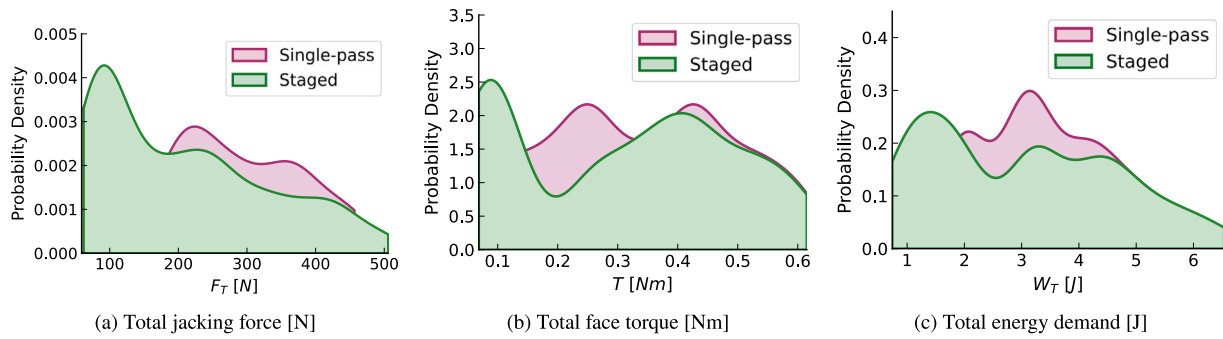


Fig. 27. Comparison of the excavation force, torque, and energy needed for the two excavation schemes.

which shows that overall, using a staged excavation method is more energy-efficient than a single-pass one.

4.3.2. Ranking of geotechnical features by importance to the excavation energy

Fig. 28 is obtained from the feature selection algorithm described in and shows the importance level of each input parameter for the estimated excavation force, torque, and total energy. In relation to the plastic shear strain descriptors, the depth-to-diameter ratio and the apparent cohesion remain important parameters in controlling the energy expenditure metrics because they control the magnitude of resisting stresses and the shear strength of the soil, respectively. The other parameters such as E , ϕ and ψ that controlled the plastic shear strain distribution and localization are not as relevant for the energy expenditure metrics due to their limited effect on stress magnitude, redistribution and shear strength as compared to D/d and c' . The depth-to-diameter ratio is the parameter that most influences the total jacking force (see Fig. 28(a)). This is because the jacking forces are calculated based on the force needed to support the lateral earth pressure at the face and the force needed to overcome friction at the sides. In the case of a single-pass excavation method, the face force is a simple function of the depth and unit weight of the soil that can be used to analytically calculate the in-situ lateral earth pressure. Side friction is a function of the machine/soil interface friction properties, and also a function of the depth and unit weight that influence the radial stress at the interface — which can also be calculated analytically for the single-pass excavation method. For these reasons, the three input parameters that most influence the total jacking force in a single-pass excavation are D/d , γ , and ϕ . For staged excavations, apparent cohesion c' emerges as the second most important parameter for the total jacking force, after D/d , and before γ and ϕ . The importance of apparent cohesion can be explained by the repeated occurrence of plasticity (at each excavation stage), which leads to a redistribution of stress, yielding a stress field with much smaller magnitudes than the original in-situ stress state. For this reason, both the radial stress and face pressure contributing to the face supporting force and the frictional force, respectively, are reduced. An instance of a simulation with low apparent cohesion that results in reduced stresses has been presented in Section 3.2.2 (Figs. 11 to 13).

The torque is affected by the variables that affect the shear strength of the soil at the excavation face. Shear strength is a function of soil apparent cohesion, internal friction angle, and the normal stress at the tunnel face. This is confirmed by Fig. 28(b), in which apparent cohesion emerges as the most important parameter in both methods due to its direct contribution to the shear strength and normal stress distribution. The next most important parameters are the depth-to-diameter ratio and unit weight, due to their influence on the stress level at the face. Interestingly, although the friction angle is the next parameter in the ranking, the importance coefficient is very small, likely due to the more significant contributions of the other parameters. Additionally, the range of values used for the friction angle — restricted to values typical of loose unsaturated sands — is not large enough to change the

shear strength significantly as compared to the apparent cohesion value and stress levels used in this study.

The importance ranks of the input parameters for the total excavation work done, shown in Fig. 28, is a combination of the importance for the force and the torque. Similar to the total jacking force, we observe an increased importance of the apparent cohesion value in the staged excavation method due to the repeated stress redistribution.

4.3.3. Effect of soil type on the excavation energy required with both excavation methods

Figs. 29 to 31 are one-to-one agreement plots that compare the two excavation methods (see Section 4.2.3 for detailed explanation of what the plots represent), with scatter points grouped by influential input parameters (c' and D/d). Fig. 29 shows that the staged excavation method requires less total jacking force for low apparent cohesion soils. Additionally, the deviation from the equivalence line is greater as the depth to diameter ratio increases. This suggests that staged excavation can reduce jacking forces in soil with low apparent cohesion and that reduction is greater for deeper cavities. Additionally, it is notable that some of the simulations appear slightly above the equivalence line which means that the jacking force is increased in these cases. As described in Section 4.3.1, this is due to the increased cumulative surface area over which the frictional resistance is applied in the staged excavation method. As expected, this is more prominent in the soils of higher shear strength where the stress redistribution is not significant enough to reduce the radial stress to the point that it compensates for the increased surface area and results in an overall reduction of frictional force. However, it is also worth noting that the deviation from the equivalence line for such cases (points below the equivalence line) is not as significant as the remaining cases (points below the equivalence line) which show a reduction in total force.

In Fig. 30, it is observed that the torque magnitude is small when the apparent cohesion value is small. Additionally, the torque required to fail the soil at the tunnel face is significantly lower with the staged excavation method than in the single-pass one for low apparent cohesion cases. The deviation from the equivalence line between the two excavation methods in these cases increases with the depth-to-diameter ratio. Overall, the observations are the same as for the jacking force: the reduction of torque estimates associated with the staged excavation is more significant for materials with low apparent cohesion and for cavities being excavated at greater depth. It is worth noting that the torque measures always lie on or below the equivalence line which shows that the torque magnitude is either unaffected or lowered by switching to the staged excavation method. This is expected because staging creates stress redistribution that reduces the shear strength that must be overcome by the torque while the cumulative area over which this stress is integrated in the staged excavation method is the same as that of the single-pass excavation method.

Both excavation methods are compared in terms of total excavation work done in Fig. 31. As expected from the observations made for the jacking force and total torque, the staged excavation method requires

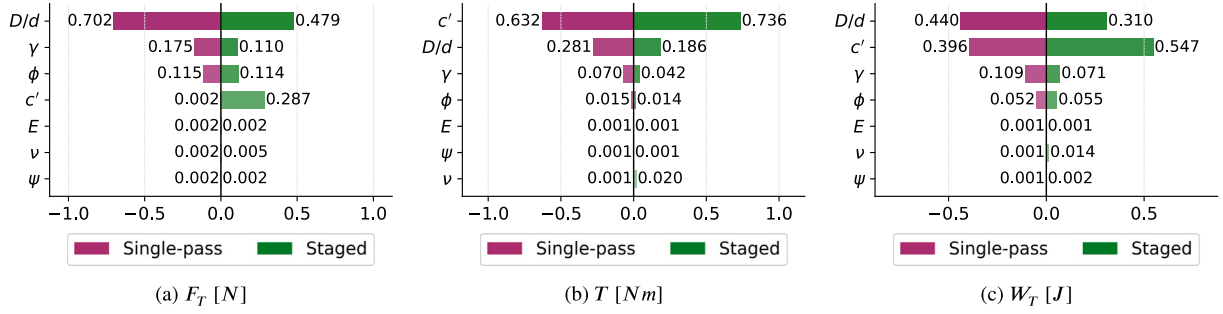


Fig. 28. Feature importance ranking for F_T [N], T [Nm], and W_T [J] in both excavation methods.

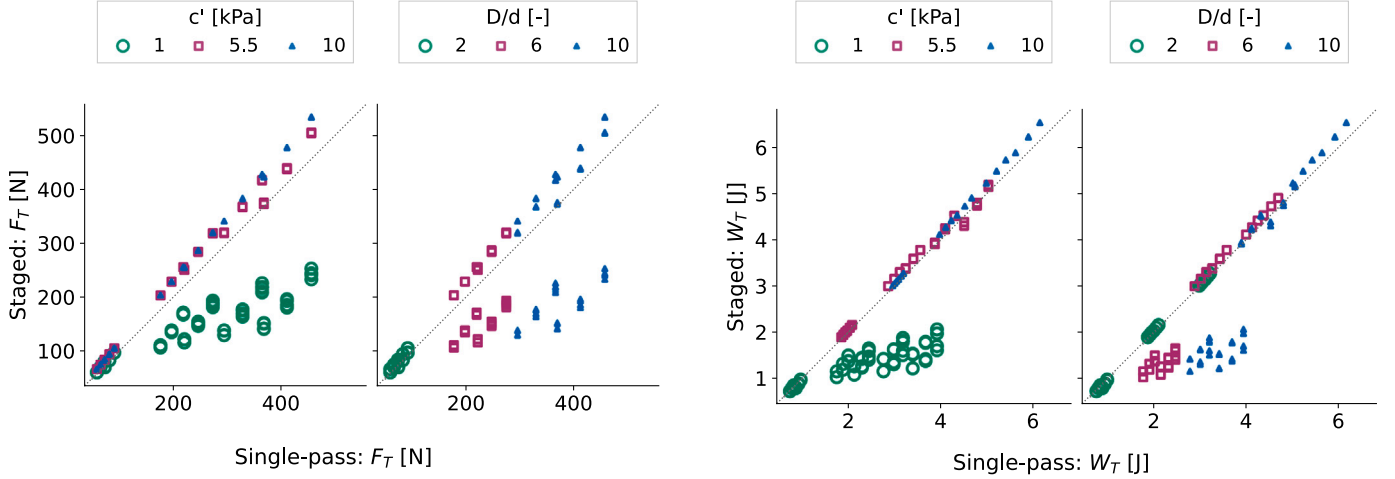


Fig. 29. Values of F_T obtained with both excavation methods, grouped by influential model parameters.

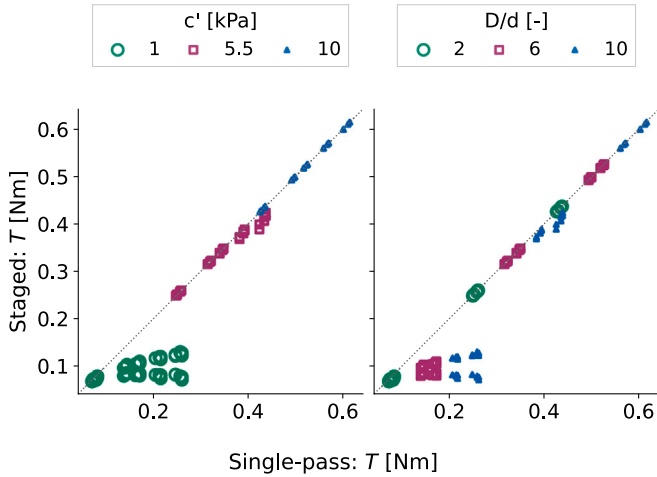


Fig. 30. Values of T obtained with both excavation methods, grouped by influential model parameters.

less energy than the single-pass one for low apparent cohesion soils, and the magnitude of this energy reduction is directly proportional to the depth-to-diameter ratio.

Table 3 provides a quantitative assessment of the reduction in force, torque and work required with the staged excavation method for the cases that significantly deviate from the equivalence line in Figs. 29–31. Specifically, cases presented in Table 3 are those where $c' = 1$ kPa and $D/d = 6$ or 10. The total jacking force is reduced by 21 to 47% with an average of 34% for an intermediate depth-to-diameter ratio of 6, while

Fig. 31. Values of W_T obtained with both excavation methods, grouped by influential model parameters.

it is reduced by 38 to 61% with an average of 50% for $D/d = 10$. For the face torque, we observe similar trends of reduction by 29 to 53% with an average of 40% for intermediate depth-to-diameter ratios and 43 to 72% with an average of 56% for high depth-to-diameter ratios. Energy savings follow the same trends since the total excavation energy represents work done by the jacking force and the torque: for $D/d = 6$, the staged excavation method yields a decrease in energy of 25 to 49% with an average of 37%, while for $D/d = 10$, an even greater reduction of 41 to 64% with an average of 53% is obtained. These results show that the staged excavation method reduces the required force, torque, and total energy required to excavate the tunnel.

5. Conclusion

Ants have long been regarded as exceptional excavators, because of their ability to carry several times their own weight, burrow tunnels with a fraction of the energy that would be needed by tunnelling boring machines per unit of excavated mass, and produce exceptionally stable excavated structures even in soils with low apparent cohesion. Inspired by the mechanical performance and efficiency of ant tunnelling, this study aims to investigate the potential mechanical benefits of the excavation technique adopted by Harvester ants, who dig tunnels in multiple passes, progressively increasing the diameter of shafts during each pass. The geometry of the excavation-induced plastic zone and the required excavation force, torque, and energy are analysed for a single-pass and a five-stage excavation process. Tunnels of diameter 10 cm have been simulated in this study which is directly relevant for small exploration robots and microtunnelling devices used to install small conduits and pipes. It can also provide insights that may be applicable for tunnels of larger diameters in more cohesive and stable soil deposits.

Table 3

Percentage of decrease in excavation force, torque and energy when using the staged excavation method over the single-pass excavation method, for cases with $c' = 1$ kPa with varying D/d : (6 or 10).

D/d	Value	F_T : $\frac{\text{single-pass} - \text{staged}}{\text{single-pass}}$	T : $\frac{\text{single-pass} - \text{staged}}{\text{single-pass}}$	W_T : $\frac{\text{single-pass} - \text{staged}}{\text{single-pass}}$
6	Min	21%	29%	25%
	Max	47%	53%	49%
	Mean	34%	40%	37%
10	Min	38%	43%	41%
	Max	61%	72%	64%
	Mean	50%	56%	53%

Two-dimensional Finite Element simulations are conducted for over 640 combinations of soil elasto-plastic parameters and cavity depth-to-diameter ratios to calculate statistical distributions of geometric descriptors of the plastic zone. Stress magnitudes calculated with the Finite Element Method are introduced in analytical expressions to estimate the total jacking force, torque, and energy required to excavate an elementary length of tunnel.

In this study, the depth-to-diameter ratio D/d and soil apparent cohesion c' are the parameters that most influence the plastic zone and the required excavation energy in both the single-pass and staged excavation processes. The Young's modulus E is found important in both schemes, but only for the plastic zone — not for energy costs. The influence of D/d is more pronounced for single-pass excavations, while the influence of c' is stronger for staged excavations. D/d translates the effect of the overburden stress: for the single-pass excavation scheme, an increase in overburden stress increases lateral earth pressure and radial stresses around the cavity induces a significant increase in thrust and energy demand. For the staged excavation method, c' becomes more important than the depth-to-diameter ratio due to greater stress redistribution associated with lower apparent cohesion. For a staged excavation, the energy demand increases significantly with increasing soil apparent cohesion c' due to the lower plasticity and stress redistribution combined with higher shear strength.

Both excavation schemes yield plastic zones that have similar, Euclidian, contour shapes for a majority of the simulations. In staged excavations, incremental mass removal induces stress redistributions and localized plastic zones that concentrate in the form of shear bands for some cases. For a single-pass excavation, plastic shear deformation is more uniformly distributed around the cavity. As a result, the shape of the plastic zone that results from a staged excavation is more complex than that from a single-pass excavation: it is typically not compact, presents a hierarchy of finer, intersecting shear bands, and exhibits tortuous boundaries. These geometric complexity characteristics are rigorously measured through fractal dimensions and Minkowski functionals. D/d , c' , ϕ , ψ and E are influential parameters of the plastic zone in both excavation schemes with ψ being a more controlling parameter for some of the plastic shear strain characteristic metrics in the staged excavation method. This is because the dilation angle controls the strain concentration along the shear bands that are formed due to a staged excavation method. The effect of the friction angle on the plastic shear strain magnitudes is more pronounced in the staged excavation process, which highlights the importance of soil strength on the intensity of localized plastic strains. A more refined analysis on the influence of soil type on the geometry of the plastic zone confirms that a staged excavation process leads to more localized plastic deformation zones for soils with a combination of low apparent cohesion, higher internal friction angle, higher stress levels and lower dilation angles. The complexity of the plastic zone induced by staged excavation is enhanced when the tunnel is at greater depth ($D/d = 6$ and $D/d = 10$).

In some cases that were simulated, the likelihood of needing high excavation force, torque, and energy is higher for a single-pass excavation than for a staged excavation, even when the force, torque, and energy components are summed over all the excavation passes. This fact is attributed to incremental stress redistribution and arching,

which occur after each excavation stage. This lowers the lateral earth pressure that must be supported at the excavation front and the radial stress that controls the interface friction on the sides of the tunnelling machine. This means that for the low apparent cohesion and greater depth cases explored in this study, excavation requires less power and less energy if staged, which has important implications for the type of engine that can be used for the boring machine. Depth-to-diameter ratio D/d , the soil specific weight γ , and the friction angle ϕ are strongly correlated to the total jacking force, the total torque and the energy demand, due to their relation to the overburden stress (for D/d and γ) or internal and interface soil strength (for ϕ). For staged excavations, the apparent cohesion c' also influences the jacking force, the torque, and the energy demand, because the efficiency of the staged excavation process depends on the formation of localized plastic zones, which are more prominent when soil apparent cohesion is low. The staged excavation method requires less total jacking force, less torque, and therefore less energy for low apparent cohesion soils, particularly for deeper cavities. At sufficiently high apparent cohesion, plasticity and arching effects are hindered, diminishing the energy efficiency of staging.

In summary, this study demonstrates that emulating the staged excavation process observed in harvester ants can significantly reduce the force, torque, power, and energy required for microtunnelling compared to traditional single-pass excavation methods. These energy gains are most pronounced in low apparent cohesion soils and at greater depth-to-diameter ratios — conditions that are characteristic of harvester ant nests excavated in loose, unsaturated sands. The underlying mechanism responsible for these reductions is the incremental development of localized plastic zones, which promotes stress redistribution and arching. This process reduces the lateral earth pressures acting at the excavation front, the frictional resistance mobilized along the tunnel perimeter, and the shearing torque required at the face. The stabilizing role of arching in ant nest excavation has been demonstrated in prior work (Frost et al., 2017; Belachew et al., 2024, 2025); here, we further show that controlled plasticity can be advantageous for tunnel construction and operation by lowering excavation energy demands.

Importantly, the proposed bio-inspired mechanism is not restricted to small-scale tunnels and dimensional analysis can be used to assess how the model parameters need to scale as the dimensions of the problem change. The observed stress and strain redistribution mechanisms are preserved at larger scales provided that similitude is maintained. If geometric dimensions are scaled by a factor λ while density remains unchanged, geostatic stresses scale linearly with λ . Maintaining the same dimensionless strain magnitudes in the scaled model therefore requires a corresponding scaling of elastic stiffness, such that the elastic modulus scales with stress ($\epsilon = \frac{\sigma}{E} = \frac{\sigma'}{E'} = \frac{\lambda\sigma}{\lambda E} \implies E' = \lambda E$), while Poisson's ratio (also dimensionless) remains unchanged. Similarly, within a Mohr–Coulomb framework, preserving frictional behaviour — which is governed by dimensionless parameters — requires cohesive strength to scale proportionally with stress ($\tan \phi = \frac{\tau - c}{\sigma} = \frac{\tau' - c'}{\sigma'} = \frac{\lambda\tau - \lambda c}{\lambda\sigma} \implies c' = \lambda c$) while the dimensionless coefficient of friction remains unchanged. Under these conditions, the stress paths, plasticity development, and arching behaviour remain mechanically similar across scales.

This above dimensional analysis suggests that the benefits of staged, ant-like excavation may extend beyond small scale tunnels, motivating future studies focused on field applications, optimization of excavation parameters, and the design of bio-inspired tunnelling and penetration devices that can achieve energy-efficient excavation.

CRedit authorship contribution statement

Meron Belachew: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yulong Liu:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **J. David Frost:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Chloé Arson:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.tust.2026.107501>.

Data availability

Data will be made available on request.

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